

Chapter 1

Introduction to Signals and Systems

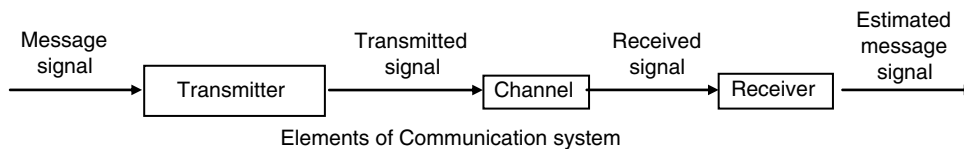
CHAPTER HIGHLIGHTS

- Classification of Signals
- Periodic and Non-periodic Signals
- Basic Operations on Signals
- Operations Performed on the Independent Variable
- Elementary Signals
- Systems and Classification of Systems
- Linear Time Invariant Systems
- Energy Signals and Power Signals
- Properties of Continuous Time LTI Systems
- Discrete Time LTI Systems

A signal is defined as a function of one or more variables that conveys information on the nature of a physical phenomenon.

When the function depends on a single variable, the signal is said to be one dimensional, (e.g., speech signal);

otherwise, it is said to be multidimensional (e.g., image). A system is defined as an entity that manipulates one or more signals to accomplish a function, thereby yielding new signals (e.g., communication system).



The transmitter changes the message signal into a form suitable for transmission over the channel.

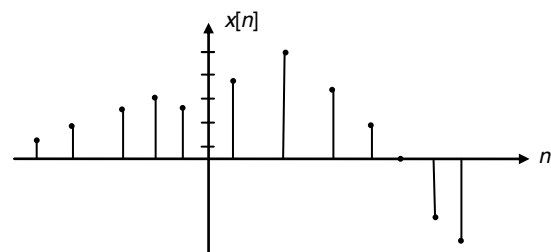
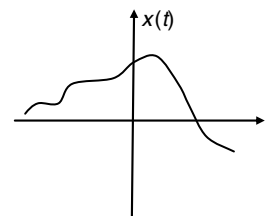
The receiver process the channel output to produce an estimate of the message signal.

CLASSIFICATION OF SIGNALS

Continuous Time and Discrete-time Signals

A signal $x(t)$ is said to be a continuous time signal, if it is defined for all time t .

The waveform in the above figure represents an example of a continuous time signal whose amplitude or value varies continuously with time.



A discrete-time signal is defined only at discrete instants of time. Thus, the independent variable has discrete values only, which are usually uniformly spaced. A discrete-time signal is often derived from a continuous time signal by sampling it at a uniform rate.

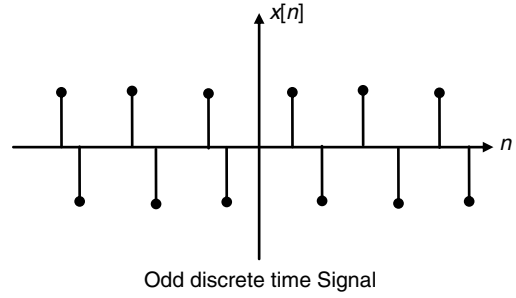
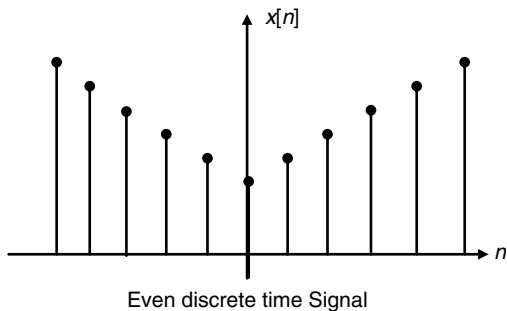
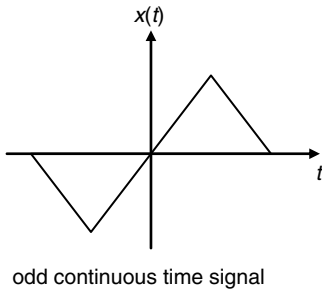
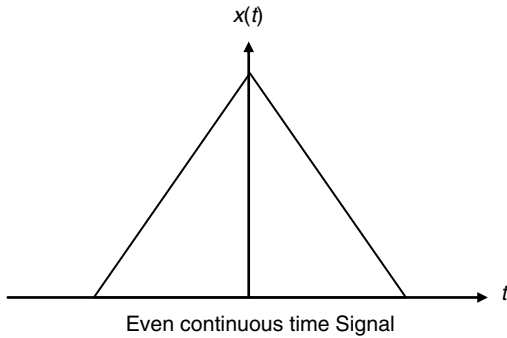
Let T_s denotes the sampling period and n denotes an integer. Then, sampling a continuous time signal $x(t)$ at time $t = nT_s$ yields a sample with values $x(nT_s)$, we can write $x[n] = x(nT_s)$, $n = 0, \pm 1, \pm 2, \dots$

Even and Odd Signals

A continuous time signal $x(t)$ is said to be an even signal if $x(-t) = x(t)$ for all t . The signal $x(t)$ is said to be an odd signal if $x(-t) = -x(t)$ for all t . A discrete-time signal $x[n]$ is even if $x[-n] = x[n]$ and $x[n]$ is odd if $x[-n] = -x[n]$.

The examples of even signal $x(t) = \cos t$, and odd signal $x(t) = \sin t$.

An odd signal must necessarily be 0 at $t = 0$ or $n = 0$; since $x(-t) = -x(t)$ or $x[-n] = -x[n]$, and therefore, $x(0) = 0$ or $x[0] = 0$.



Even signals are symmetric about vertical axis or time origin. While odd signals are anti-symmetrical about the time origin (same in the case of discrete-time signals). Any signal $x(t)$ can be expressed as sum of even and odd components.

$$x(t) = x_e(t) + x_o(t)$$

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)]$$

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)]$$

In the case of complex-valued signal, the signal $x(t)$ is said to be conjugate symmetric, if $x(-t) = x^*(t)$, where $x^*(t)$ is complex conjugate.

$$\text{If } x(t) = a(t) + jb(t)$$

$$x(-t) = a(-t) + jb(-t)$$

$$x^*(t) = a(t) - jb(t)$$

By comparing the abovementioned equations, we can say that a complex-valued signal $x(t)$ is conjugate symmetric if its real part is even [$a(t) = a(-t)$], and its imaginary part is odd [$b(-t) = -b(t)$]. (A similar remark applies for discrete signals too.)

Example 1

Find the even and odd components of the signal $x(t) = e^{-2t} \sin t$.

Solution

By replacing t with $-t$, $x(-t) = e^{2t} \sin(-t) = -e^{2t} \sin(t)$

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)] = \frac{1}{2} [e^{-2t} \sin t - e^{+2t} \sin t]$$

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)] = \frac{1}{2} [e^{-2t} \sin t + e^{2t} \sin t]$$

$$x_e(t) = \sin t \frac{1}{2} [e^{-2t} - e^{2t}] = -\sinh(2t) \sin t$$

$$x_o(t) = \sin t \frac{1}{2} [e^{-2t} + e^{2t}] = \cosh(2t) \sin t$$

Periodic and Non-periodic Signals

If $x(t) = x(t + T)$, for all t , where T is a positive constant, then $x(t)$ is said to be periodic.

If this condition satisfies for $T = T_0$, say, then it is also satisfied for $T = 2T_0, 3T_0, 4T_0, \dots$

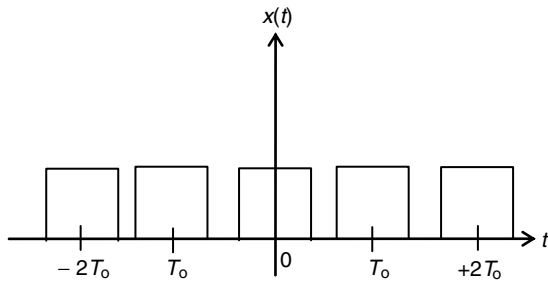
The smallest value of T that satisfies $x(t) = x(t + T)$ is called fundamental period of T and its reciprocal $f = \frac{1}{T_0}$ is called fundamental frequency.

Frequency measured in Hertz (Hz) and the angular frequency measured in radians per second is defined by $\omega = 2\pi f = \frac{2\pi}{T_0}$

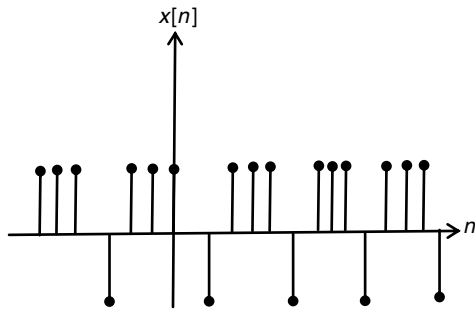
A discrete-time signal $x[n]$ is periodic with period N (where N is a positive integer), if it is unchanged by a time shift of N .

$$x[n] = x[n + N] \text{ for integer } n, N$$

Further, $x[n]$ is also periodic with period $2N, 3N, \dots$. The fundamental period N_0 is the smallest positive value of N for which of the abovementioned equation holds.



Continuous time periodic signal with T_0 .



Discrete time periodic signal with $N_0 = 4$

A signal $x(t)/x[n]$ that is not periodic will be referred to as an aperiodic signal.

Deterministic Signals and Random Signals

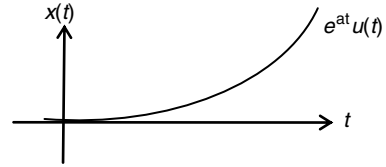
The signal that can be completely specified by a mathematical equation is called deterministic signal, otherwise random signal.

Examples of deterministic signals.

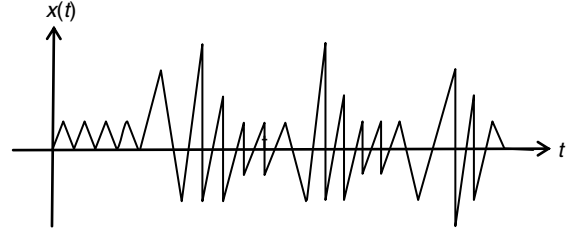
$$x(t) = 2 - 2tu(t), x(t) = \sin \omega t$$

$$x[n] = \cos \omega n, x[n] = a^n u[n]$$

A random signal is a signal about which there is uncertainty before it occurs. Such random signals are generally characterized by their mean and mean square values.



Deterministic signal



Random signal

Examples of random signals: the electrical noise generated in the amplifier and interference component in television receiver.

Energy Signals and Power Signals

Total energy of the continuous time signal

$$\begin{aligned} x(t) \text{ is } E &= \lim_{t \rightarrow \infty} \int_{-T/2}^{T/2} x^2(t) dt \\ &= \int_{-\infty}^{\infty} x^2(t) dt \end{aligned}$$

Further, its time average power

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

The time averaged power of a periodic signal $x(t)$ of fundamental period T is given by

$$P = \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

In the case of discrete-time signals, the integrals are replaced by the corresponding sums.

$$E = \sum_{n=-\infty}^{\infty} x^2[n]$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x^2[n]$$

The average power of periodic signal $x(n)$ with fundamental period N is

$$P = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n]$$

A signal is referred to as an energy signal, if and only if the total energy of the signal satisfies the condition $0 < E < \infty$.

A signal is said to be as a power signal if and only if the average power of the signal satisfies the condition.

$$0 < P < \infty.$$

The energy and power clarification of signals are mutually exclusive. An energy signal has zero time-averaged power, whereas a power signal has infinite energy.

It is of interest to note that periodic signals and random signals are usually viewed as power signals, whereas signals that are both deterministic and non-periodic are usually viewed as energy signals.

Causal, Non-causal, and Anti-causal Signals

A signal is said to be causal if it is defined only for $t > 0$,

For example, $u(t)$, $Ae^{bt}u(t)$, $\sin \omega t u(t)$

A signal is said to be non-causal if it is defined for either $t \leq 0$ or for both $t \leq 0$ and $t > 0$

For example, Ae^{bt} , for all t , $\sin \omega t$, for all t .

When a non-causal signal is defined only for $t \leq 0$, it is called anti-causal signal.

For example, $Ae^{bt}u(-t)$, $\sin \omega t u(-t)$

Example 2

What is the fundamental period of the discrete-time signal $x[n] = e^{j(2\pi/6)n} + e^{j(3\pi/4)n}$

Solution

The first exponential fundamental period $N_1 = m \left(\frac{2\pi}{2\pi/6} \right) = 6m$ will be integer for $m = 1$, and therefore, fundamental period $N_1 = 6$.

For the second exponential, the fundamental period $N_2 = m \left(\frac{2\pi}{3\pi/4} \right) = m \cdot \frac{8}{3}$ will be integer for $m = 3$, and therefore, fundamental period, $N_2 = 8$.

The least common multiple for these fundamental periods is 24, that is, the fundamental period of $x[n]$

Example 3

Find the fundamental period (in seconds) and fundamental frequency (in rad/s) for the following signals.

- $v(t) = 20\sin 100t + 10\cos 300t + 5\sin \left(500t + \frac{\pi}{3} \right)$
- $v(t) = 3\cos 200t + 4\cos 300t + 6\sin 500t$

Solution

- $v(t) = 20\sin 100t + 10\cos 300t + 5\sin \left(500t + \frac{\pi}{3} \right)$

For the abovementioned signal, we can consider individual time periods as $T_1 = \frac{2\pi}{100}$, $T_2 = \frac{2\pi}{300}$, $T_3 = \frac{2\pi}{500}$.

The least common multiple (LCM) of T_1 , T_2 , and T_3 is $\frac{2\pi}{100} = T_0$ is the fundamental time period in seconds.

$T_0 = \frac{2\pi}{100}$, the fundamental frequency is $\omega_0 = \frac{2\pi}{T_0} = 100 \text{ rad/s}$

- $v(t) = 3\cos 200t + 4\cos 300t + 6\sin 500t$. The individual time periods are $T_1 = \frac{2\pi}{200}$, $T_2 = \frac{2\pi}{300}$, $T_3 = \frac{2\pi}{500}$

The LCM of (T_1, T_2, T_3) , $= \frac{2\pi}{100} = T_0$ is the fundamental period

[Therefore, check $v(t + T_0) = v(t)$]; fundamental frequency $(\omega_0) = \frac{2\pi}{T_0} = 100 \text{ rad/s}$.

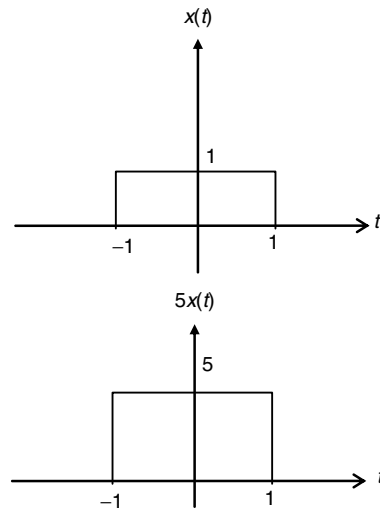
BASIC OPERATIONS ON SIGNALS (OPERATIONS PERFORMED ON DEPENDENT VARIABLES)

Amplitude Scaling

$$y(t) = Cx(t)$$

$$y[n] = Cx[n]$$

The value of $y(t)$ is obtained by multiplying the corresponding value of $x(t)$ by the scalar C for each instant of time t . A physical example of a device that performs amplitude scaling is an electronic amplifier.

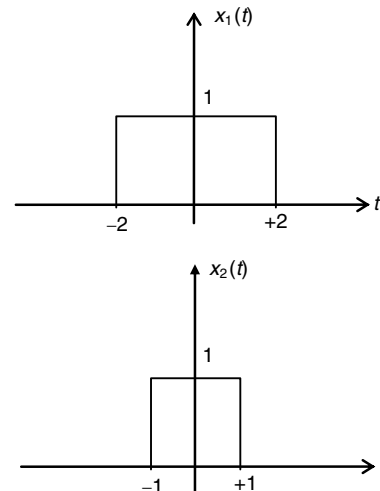


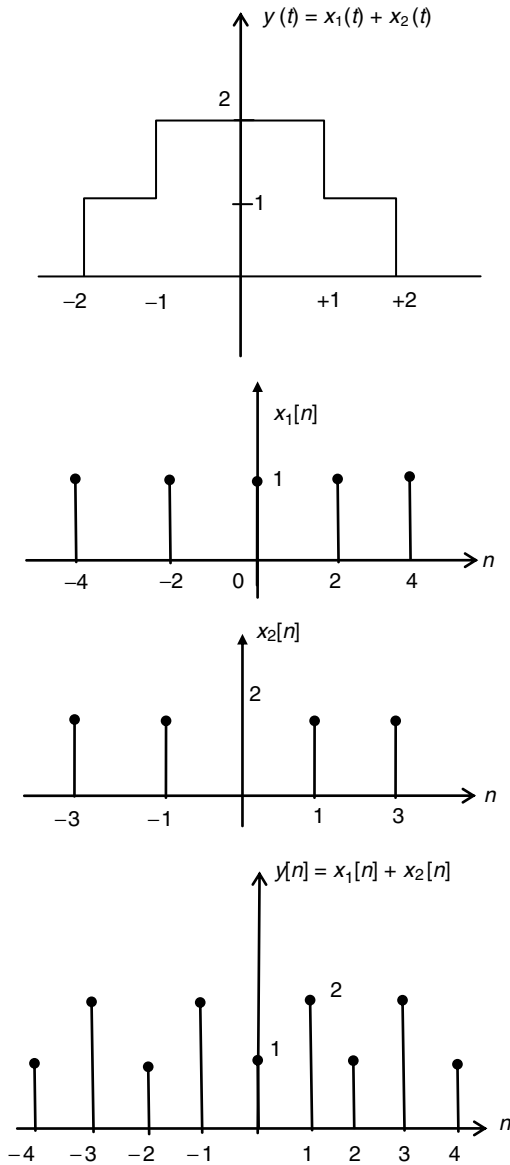
Addition

$$y(t) = x_1(t) + x_2(t)$$

$$y[n] = x_1[n] + x_2[n]$$

$x_1(t)$ and $x_2(t)$ are continuous signals, then $x_1(t) + x_2(t)$ denote addition of the two signals.





Multiplication

$$y[n] = x_1[n] \cdot x_2[n]$$

$$y(t) = x_1(t) \cdot x_2(t)$$

If $x_1(t)$ and $x_2(t)$ are two continuous time signals, then their product $y(t) = x_1(t) \cdot x_2(t)$. The value of $y(t)$ at time ' t ' is given by the product of the corresponding values of $x_1(t)$ and $x_2(t)$.

For example in AM modulated signal, the message signal will be multiplied by the carrier signal, and the resultant changes the amplitude of carrier as per message signal.

Differentiation

$$y(t) = \frac{d}{dt} x(t),$$

$$y[n] = x[n] - x[n-1]$$

The derivative of $x(t)$ with respect to time is defined as

$$y(t) = \frac{d}{dt} x(t)$$

For example, the voltage developed across inductor is given by $v(t) = L \cdot \frac{di(t)}{dt}$

Integration

$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

$$y[n] = \sum_{m=-\infty}^n x[m]$$

The integral of $x(t)$ with respect to time ' t ' is defined as

$$y(t) = \int_{-\infty}^t x(\tau) d\tau.$$

For example, the voltage across capacitor is given by

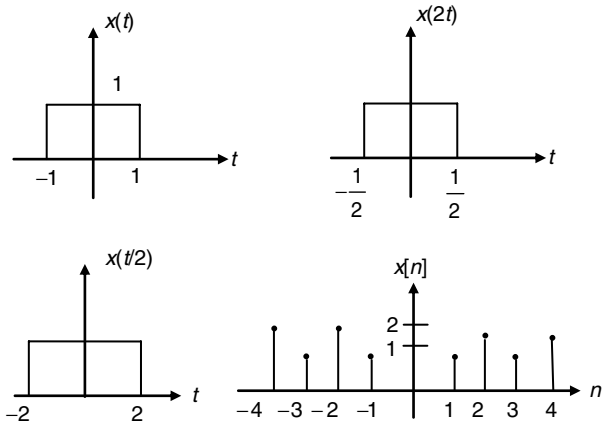
$$v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau.$$

OPERATIONS PERFORMED ON THE INDEPENDENT VARIABLE

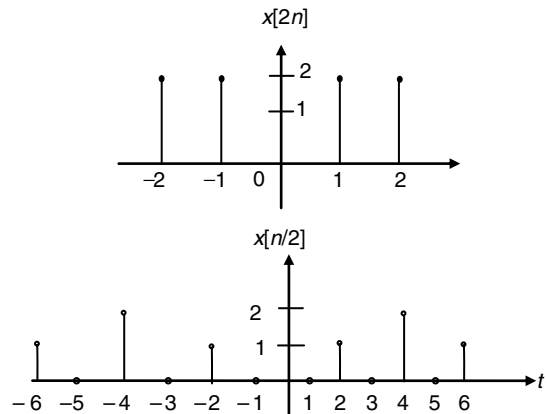
Time Scaling

$$y(t) = x(at)$$

If $a > 1$, the signal $y(t)$ is a compressed version of $x(t)$; if $0 < a < 1$, the signal $y(t)$ is an expanded (stretched) version of $x(t)$.



In this $x(t)$, $x(2t)$, $x(t/2)$ are one at whereas $x(n)$, $x(2n)$, $x(n/2)$ are another set.

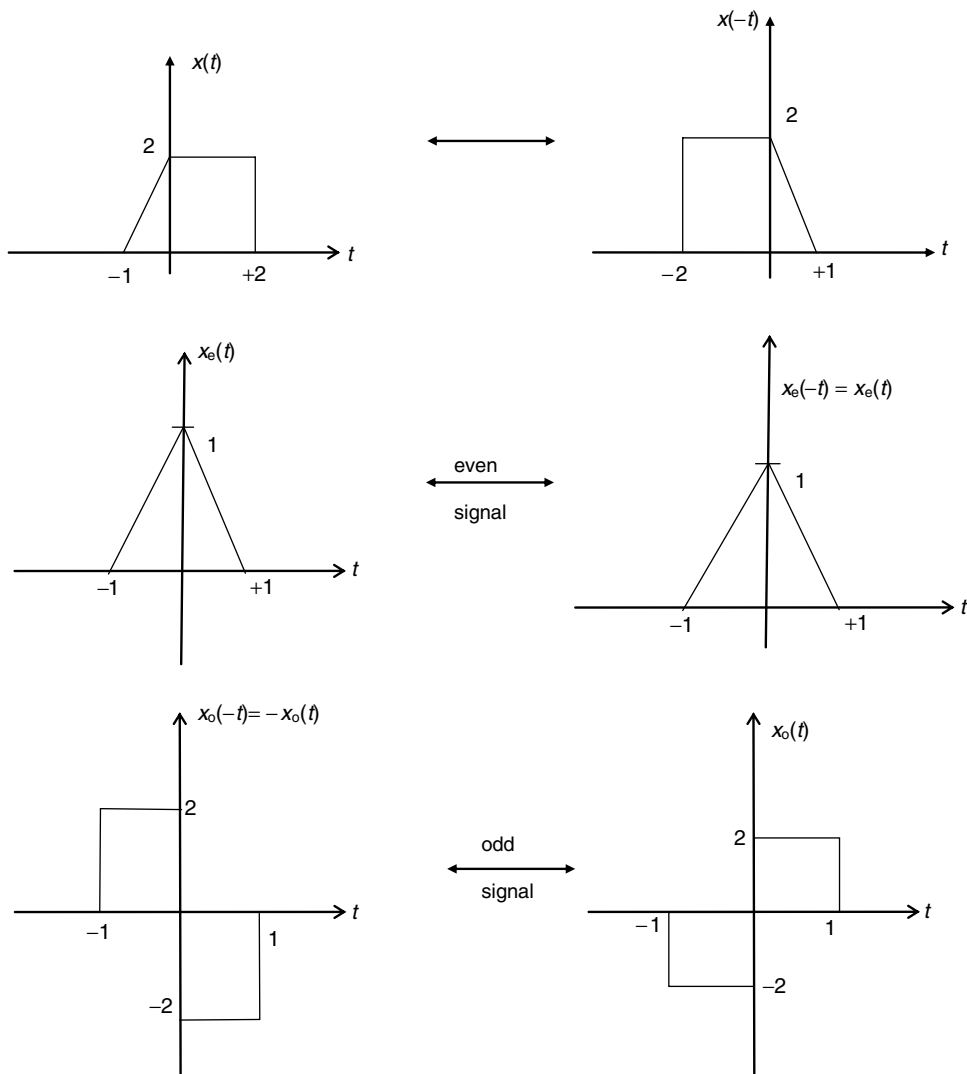


Reflection

$$y(t) = x(-t)$$

Even signal $[x(-t) = x(t)]$ is same as reflected signal; odd signal $[x(-t) = -x(t)]$ is negative of its reflection.

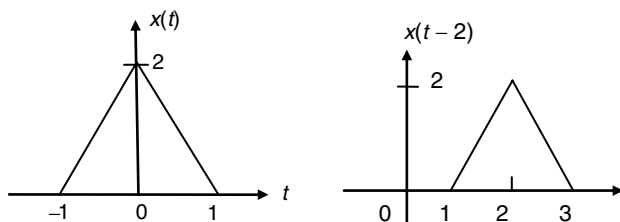
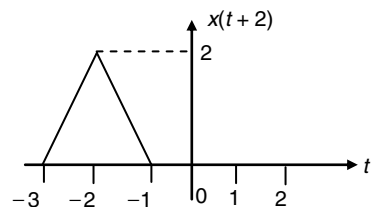
Reflected signal can be obtained by replacing t by $-t$.



Time Shifting

$$y(t) = x(t - t_0)$$

If $t_0 > 0$, the waveform of $y(t)$ is obtained by shifting $x(t)$ toward the right, relative to the time axis, if $t_0 < 0$, $x(t)$ is shifted to the left.



Precedence Rule for Time Shifting and Time Scaling

If $y(t) = x(at - b)$.

To obtain $y(t)$ from $x(t)$, the proper order is based on the fact that the scaling operation always replaces ' t ' by ' at ', while time shifting operation always replaces ' t ' by ' $t - b$ '.

Hence, time shifting operation is performed first on $x(t)$, resulting in an intermediate signal $\vartheta(t) = x(t - b)$.

Next time scaling operation is performed on $\vartheta(t)$ resulting in desired output $y(t) = \vartheta(at) = x(at - b)$.

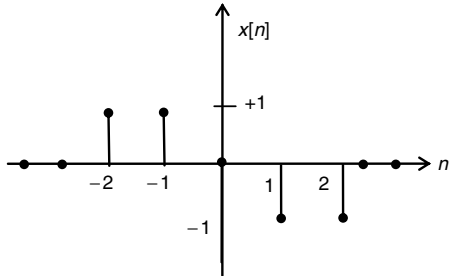
Incorrect way is first time scaling $[x(at)]$ and then applying time shifting results in $x[a(t - b)] \neq x[at - b]$. Similar approach we follow for discrete-time signals.

Example 4

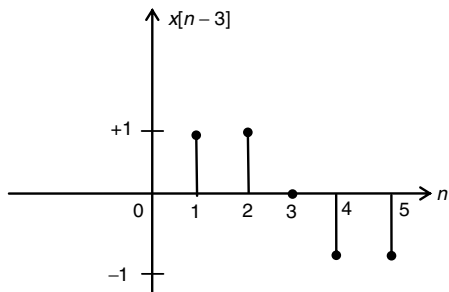
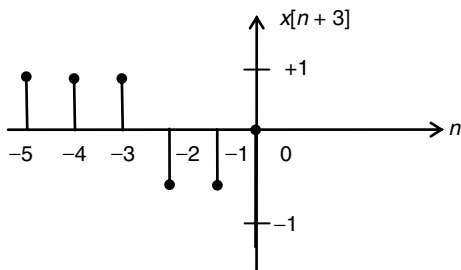
The discrete-time signal

$$x[n] = \begin{cases} 1 & n = -1, -2 \\ -1, & n = 1, 2 \\ 0, & n = 0, n > 2 \end{cases}$$

Find $x[n + 3]$ and $x[n - 3]$.

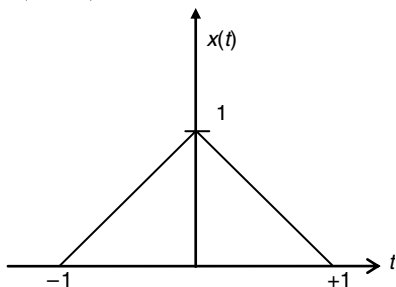


Solution

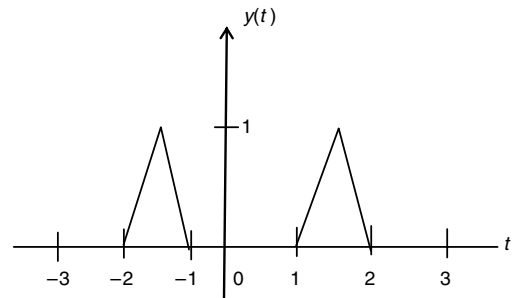
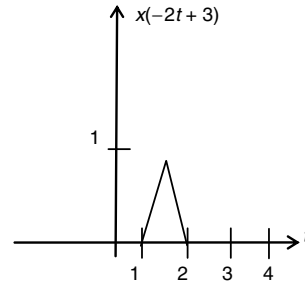
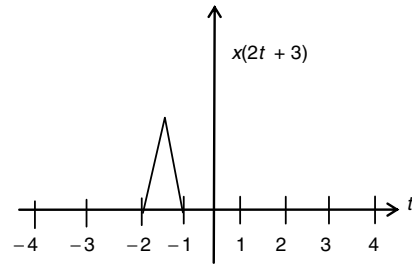
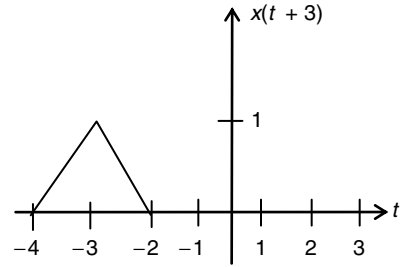


Example 5

A triangular pulse $x(t)$ is depicted in figure sketch $y(t) = x(-2t + 3) + x(2t + 3)$

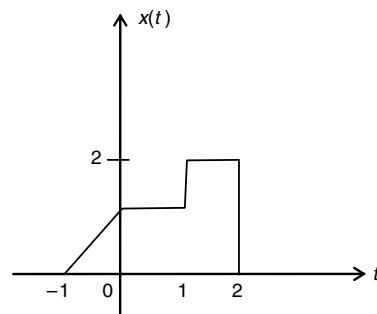


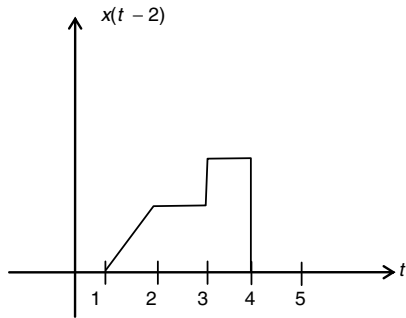
Solution



Example 6

For the signal $x(t)$ depicted in the figure, draw the signals $x(t - 2)$, $x(2t + 1)$, $x(3 - t)$, $x(-t - 1)$, $[x(-2t) + x(2t)]$?

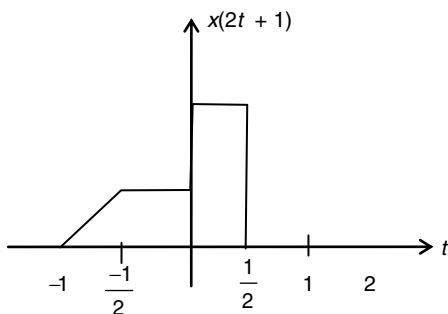
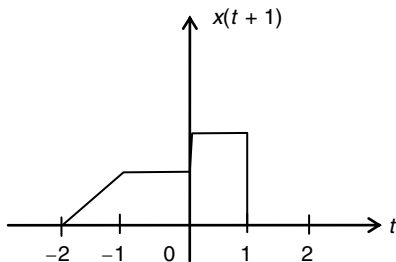


Solution

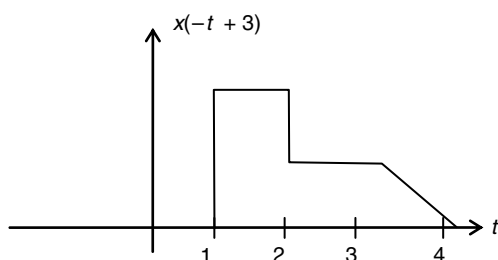
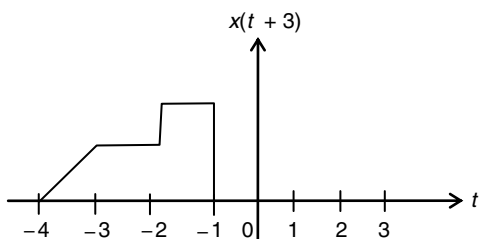
$x(t-2)$ – right shift by 2

To find out $x(2t+1)$, scaling by 2 replaces t by $2t$; similarly, shifting by +1 replaces t by $t+1$.

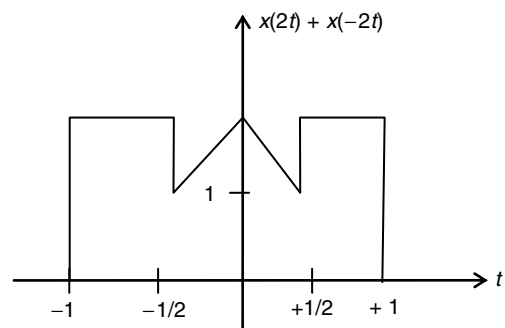
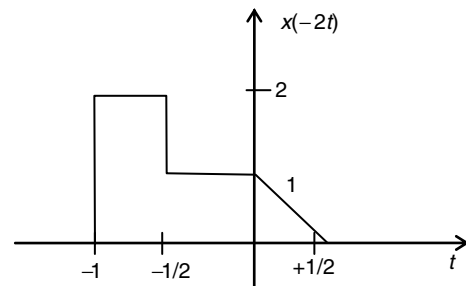
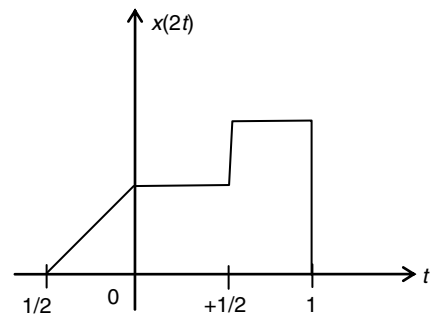
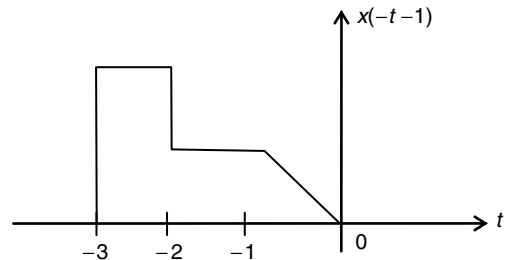
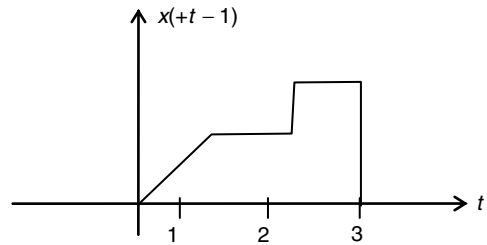
To get $x(2t+1)$, first we need to perform shifting $x(t+1)$ and then scaling $x(2t+1)$; however, first scaling results in $x(2t)$ and then shifting results in $x(2(t+1)) = x(2t+2)$, which is not desired.



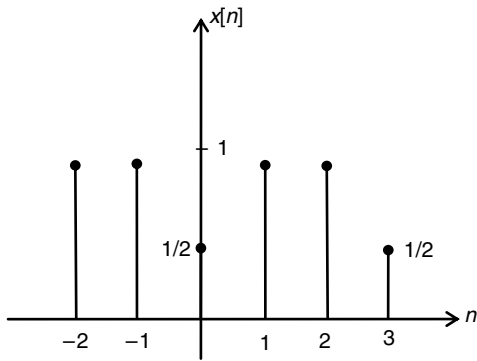
$x(3-t)$; first shifting $x(t+3)$, then scaling by -1 , that is, $x(-t+3)$



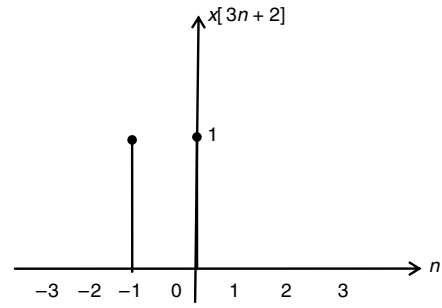
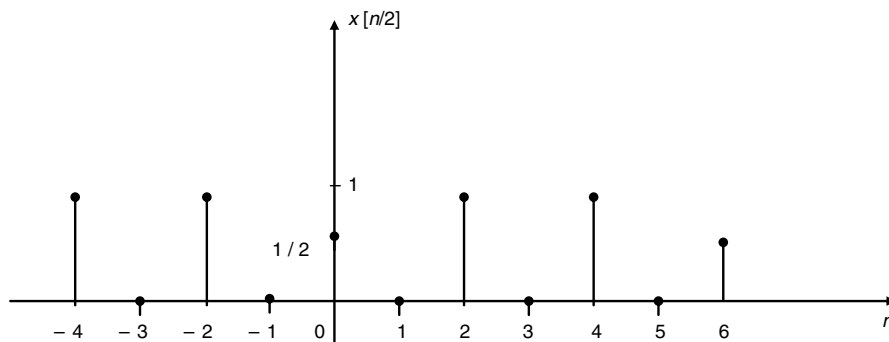
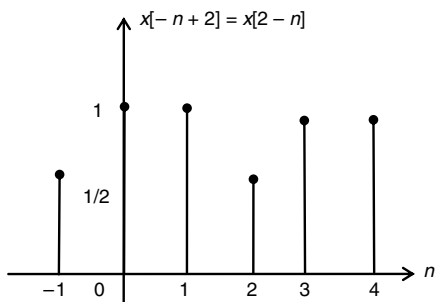
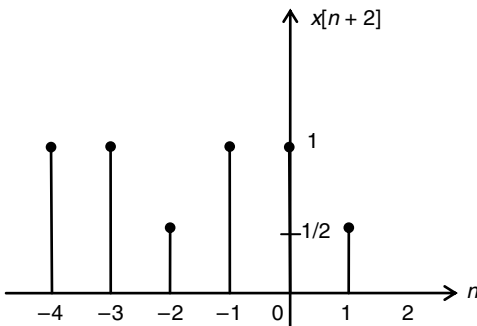
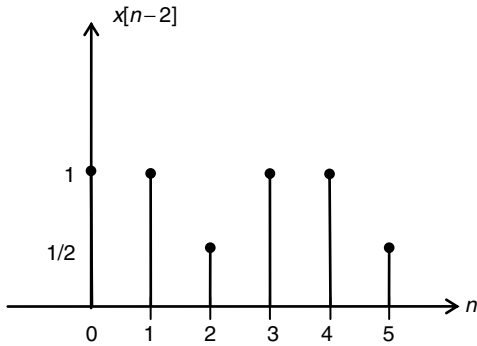
For $x(-t-1)$, first shifting $x(t-1)$, then scaling $x(-t-1)$

**Example 7**

For the discrete-time signal shown in the figure, draw the following signals $x[n-2]$, $x[2-n]$, $x[3n+2]$, $x[n/2]$, $x[n] + x[-n]$.



Solution



$x[3n+2]$ will be defined in the range of

$$-2 \leq 3n+2 \leq 3$$

$$-4 \leq 3n \leq 1$$

$$-4/3 \leq n < 0/3$$

$$-1 \leq n \leq 0$$

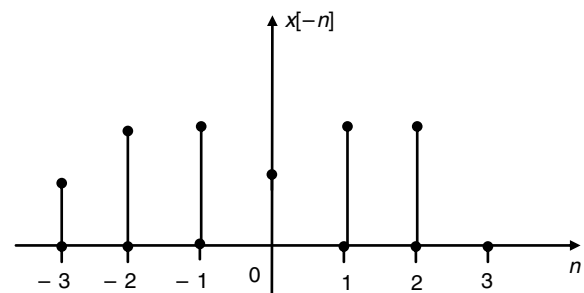
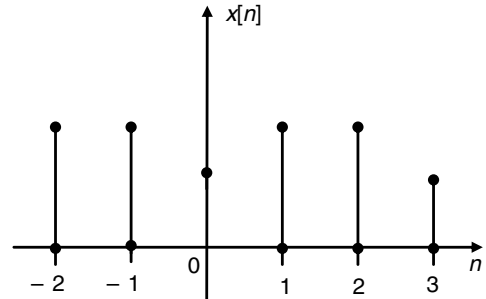
$$x(3n+2), x(3 \times 0 + 2) = x(2) = 1$$

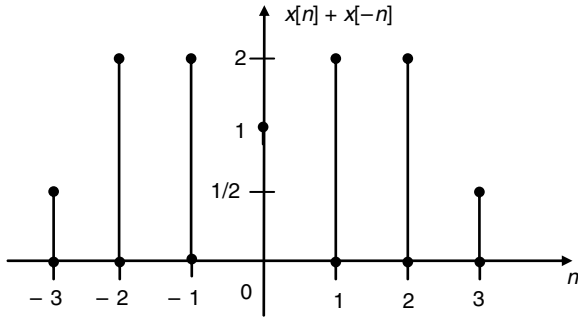
$$x(3 \times -1 + 2) = x(-1) = 1$$

For $x[n/2]$, the range is $-2 \leq n/2 \leq 3 \Rightarrow -4 \leq n \leq 6$

$x[n/2]$ for $n = 4$ is $x[4/2] = x[2] = 1$

For $n = -3$, $x[-3/2] = 0$, which is not defined.





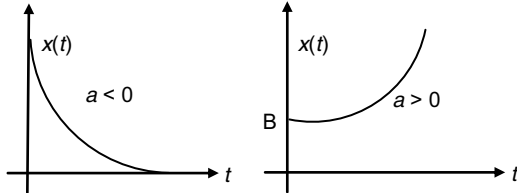
ELEMENTARY SIGNALS

Exponential Signals

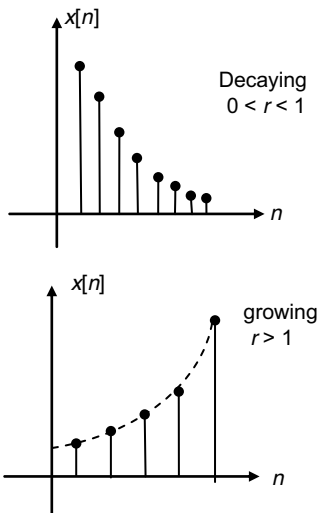
$$x(t) = Be^{at}$$

a, B are real parameters, and B is the amplitude of the exponential signal measured at $t = 0$; when $a < 0$, it is decaying exponential.

When $a > 0$, it is growing exponential.



In discrete time, real exponential signal $x[n] = Br^n$ $r = e^a$,



For $r < 0$, the discrete-time exponential $x[n]$ assumes alternating signs for the n ; r^n is positive for n even and negative for n odd.

Sinusoidal Signals

$$x(t) = A \cos(\omega t + \phi)$$

A sinusoidal signal is an example of periodic signal $x(t + T) = x(t)$

$$T = \frac{2\pi}{\omega}$$

Consider discrete-time signal

$$x[n] = A \cos(\Omega n + \phi)$$

It is periodic signal

$$x[n + N] = x[n],$$

where $N = \frac{2\pi m}{\Omega}$ and m, N integers.

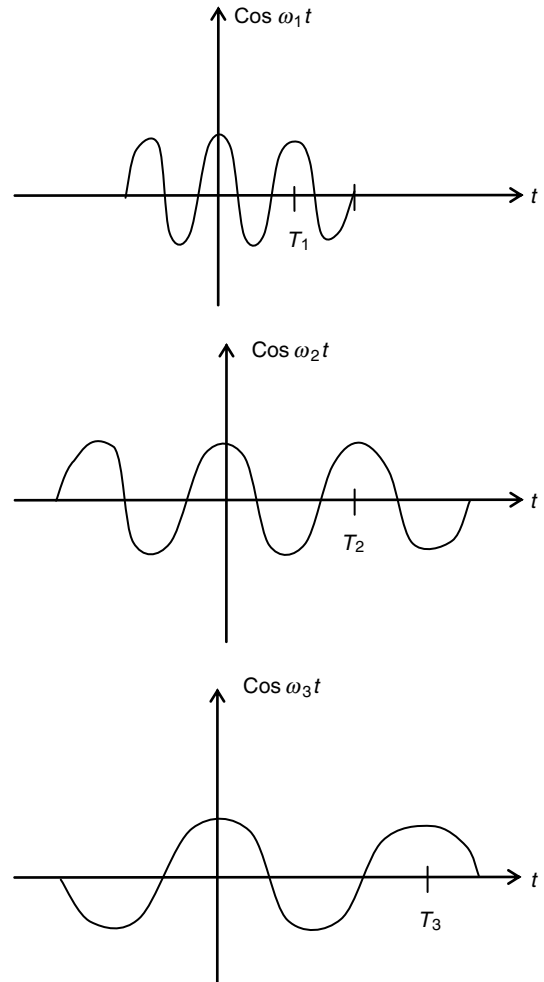
The angular frequency Ω must be a rational multiple of 2π . The relation between sinusoidal and complex exponential signals is $e^{j\theta} = \cos\theta + j\sin\theta$

$x(t) = e^{j\omega_o t}$ is a complex exponential signal. An important property of this signal is that it is periodic, $e^{j\omega_o t} = e^{j\omega_o(t+T)} = e^{j\omega_o t} e^{j\omega_o T}$ for periodicity $e^{j\omega_o T} = 1$

This is true for the fundamental period $T_o = \frac{2\pi}{|\omega_o|}$.

Thus, the signals $e^{j\omega_o t}$ and $e^{-j\omega_o t}$ have the same fundamental period.

$e^{j\omega_o t} = \cos \omega_o t + j\sin \omega_o t$ (from Euler's relations), we can observe from the relation $T_o = \frac{2\pi}{|\omega_o|}$ that fundamental period T_o is inversely proportional to the fundamental frequency ω_o .



For the above three sinusoidal, $\omega_1 > \omega_2 > \omega_3$, which implies that $T_1 < T_2 < T_3$.

In the case of discrete-time complex exponentials, $e^{j\omega_o n}$ periodic only if $\omega_o = \frac{2\pi M}{N}$ for some integers M, N the

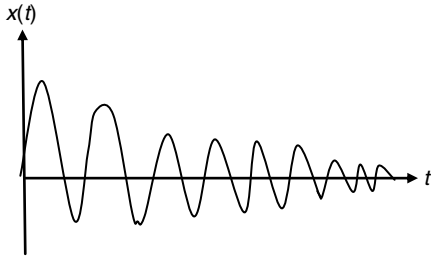
fundamental period will be $N = M \left(\frac{2\pi}{\omega_o} \right)$ for integers M, N , assuming M and N do not have any factors in common.

Exponentially Damped Sinusoidal Signals

The multiplication of a sinusoidal signal by a real-valued decaying exponential signal results in a new signal referred to as an exponentially damped sinusoidal signal.

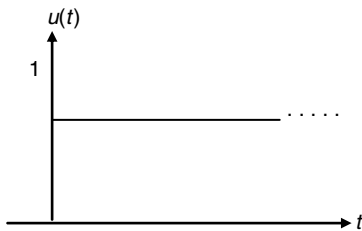
The discrete-time version of the exponentially damped sinusoidal signal is $x[n] = Ar^n \sin[\Omega n + \phi]$. For the signal to decay exponentially with time, the parameter r should lie in the range $0 < |r| < 1$.

For continuous time, $x(t) = Ae^{-\alpha t} \sin(\omega t + \phi)$ $\alpha > 0$



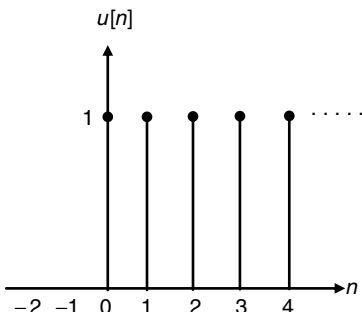
Step Function

$$\begin{aligned} u(t) &= 1, t > 0 \\ &= 0, t < 0 \\ &= 1/2, t = 0 \end{aligned}$$



For discrete-time version

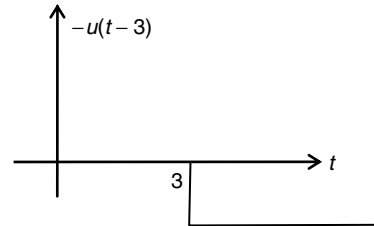
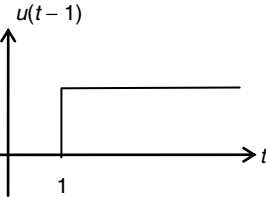
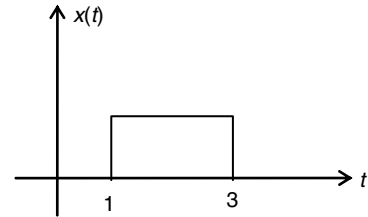
$$\begin{aligned} u[n] &= 1, n \geq 0 \\ &= 0, n < 0 \end{aligned}$$



The unit step function $u(t)$ is particularly a simple signal to apply as a test signal, and the unit step function is useful because the output of a system due to a step input reveals a great deal about how quickly the system responds to an abrupt change in the input signal. A similar remark applies to $u[n]$ in the context of discrete-time system.

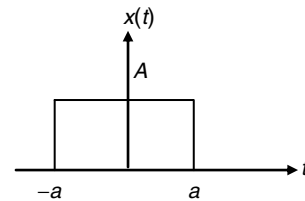
If we want a signal to start at $t = 0$ (so that it has value of 0 for $t < 0$), we need to multiply the signal by $u(t)$.

Consider a rectangular pulse from $t = 1$ to $t = 3$, and it can be derived from delayed unit impulse functions as shown here:



$$x(t) = u(t-1) - u(t-3)$$

Rectangular Pulse



$$\begin{aligned} x(t) &= A, 0 \leq |t| \leq a \\ &= 0, |t| > 0 \end{aligned}$$

It can be written as $x(t) = A u(t+a) - A u(t-a)$

Impulse Function

In discrete time

$$\delta[n] = \begin{cases} 1, n = 0 \\ 0, n \neq 0 \end{cases}$$

The continuous time version of the unit impulse is defined by $\delta(t) = 0$ for $t \neq 0$.

$$\text{Further, } \int_{-\infty}^{\infty} \delta(t) dt = 1.$$

The impulse $\delta(t)$ is also referred to as the Dirac delta function.

We can visualize an impulse function as a tall, narrow, rectangular pulse of unit area, and the width of this rectangular pulse is very small $\leftarrow \rightarrow 0$; consequently, its height is very large $\frac{1}{\Delta} \rightarrow \infty$

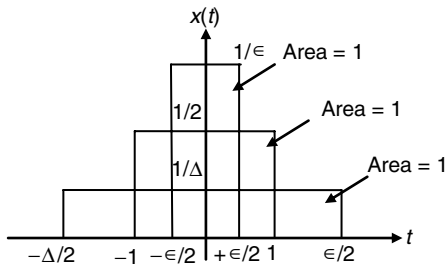


Figure 1 Evolution of a rectangular pulse of unit area into an impulse of unit strength

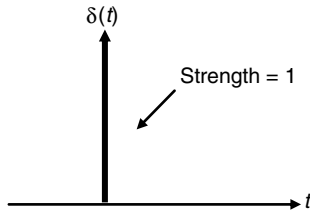


Figure 2 Unit impulse

The impulse $\delta(t)$ and unit step function $u(t)$ are related to each other as $\delta(t) = \frac{d}{dt} [u(t)]$ or $\delta[n] = u[n] - u[n-1]$

$$\text{Conversely, } u(t) = \int_{-\infty}^t \delta(\tau) d\tau = \int_0^{\infty} \delta(t-\tau) d\tau \quad (\text{or})$$

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k] = \sum_{k=-\infty}^n \delta[k]$$

The properties of impulse function.

1. $\delta(-t) = \delta(t)$
2. $x(t)\delta(t) = x(0)\delta(t)$, $x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$
3. $\int_{-\infty}^{\infty} x(t)\delta(t)dt = x(0)$, $\int_{-\infty}^{\infty} x(t)\delta(t-t_0)dt = x(t_0)$
4. $\int_{-\infty}^{\infty} x(\lambda)\delta(t-\lambda)d\lambda = x(t)$
5. $\delta(at) = \frac{1}{a}\delta(t)$, $a > 0$

$$6. \int_{-\infty}^{\infty} \delta(t)dt = 0$$

$$7. \int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt = \left. \frac{d}{dt} f(t) \right|_{t=t_0}$$

$$8. \int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt = \left. \frac{d^n}{dt^n} f(t) \right|_{t=t_0}$$

$$9. x[n]\delta[n] = x[0]\delta[n],$$

$$x[n]\delta[n-n_0] = x[n_0]\delta[n-n_0]$$

$$10. x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

$$\text{For example, } \frac{\omega^2 + 1}{\omega^2 + 9} \delta(\omega - 1) = \frac{1^2 + 1}{1^2 + 9} (\omega - 1) = \frac{1}{5} \delta(\omega - 1)$$

$$\sin\left(t - \frac{\pi}{2}\right) \delta(t) = \sin\left(0 - \frac{\pi}{2}\right) \delta(t) = -\delta(t)$$

$$\int_{-\infty}^{\infty} \delta(t-4) \cos\left(\frac{\pi t}{4}\right) dt = \cos\left(\frac{\pi}{4} \cdot 4\right) = \cos \pi = -1$$

$$\int_{-\infty}^{\infty} e^{-2(x-t)} \delta(2-t) dt = e^{-2(x-2)}$$

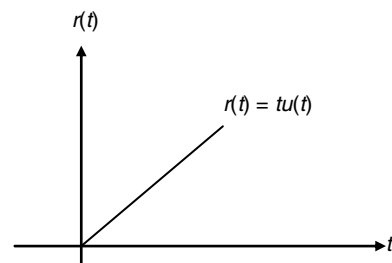
Ramp Function

The impulse function $\delta(t)$ is the derivative of the step function $u(t)$ with respect to time. By the same token, the integral of the step function $u(t)$ is a ramp function of unit slope.

$$r(t) = t, \quad t \geq 0,$$

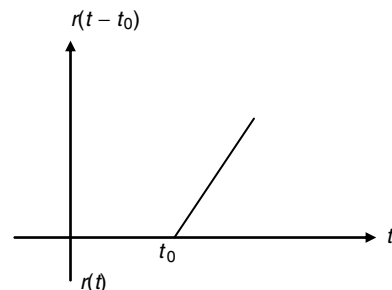
$$0, \quad t < 0$$

$$\text{or } r(t) = t u(t)$$

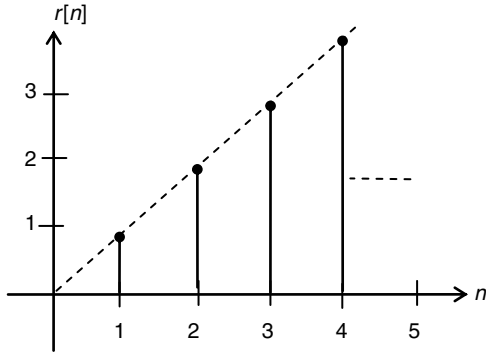


$$r(t-t_0) = t-t_0, \quad t \geq t_0$$

$$= 0, \quad t < t_0$$



The discrete-time version of the ramp function is defined by

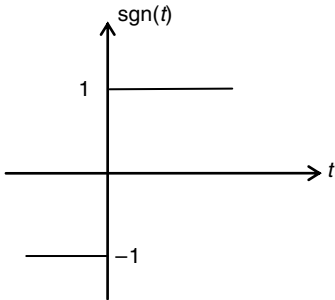


$$\begin{aligned} r[n] &= n, n \geq 0 \\ &= 0, n \leq 0 \\ \text{(or)} \quad r[n] &= nu[n] \end{aligned}$$

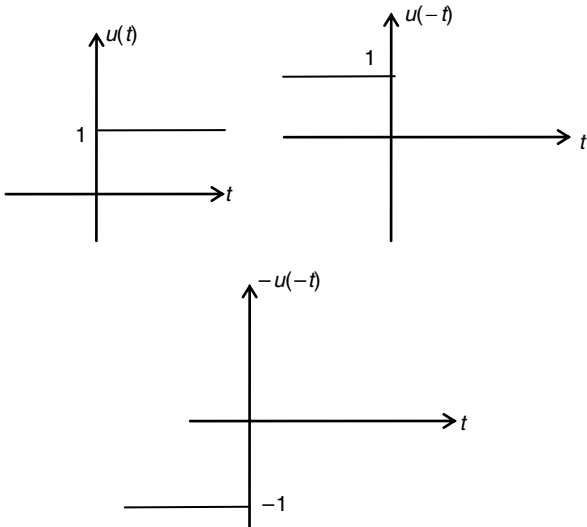
Signum Function

The signum function is defined as

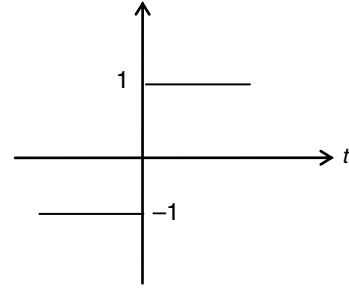
$$\text{sgn}(t) = \begin{cases} -1 & ; t < 0 \\ 0 & ; t = 0 \\ +1 & ; t > 0 \end{cases}$$



The representation is shown in the figure, and it can be derived from unit step function $u(t)$ and folded unit step function $u(-t)$.



$$\text{sgn}(t) = u(t) - u(-t)$$



$$\begin{aligned} \text{sgn}(t) &= u(t) - u(-t) = u(t) - [1 - u(t)] = 2u(t) - 1, \\ \text{In discrete time} \end{aligned}$$

$$\text{sgn}[n] = \begin{cases} -1 & , n < 0 \\ 0 & , n = 0 \\ +1 & , n > 0 \end{cases}$$

$$\text{sgn}[n] = u[n] - u[-n] = 2u[n] - 1$$

Relation between Elementary Signals

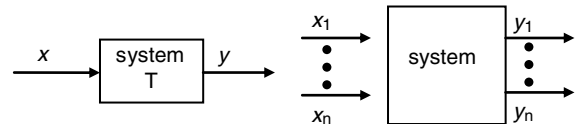
1. $\frac{d}{dt} u(t) = \delta(t)$, or $u(t) = \int_0^{\infty} \delta(t - \sigma) d\sigma$
2. $x(t)u(t) = 0$ for $t < 0 = x(t)$ for $t \geq 0$
3. $\frac{d}{dt} r(t) = u(t)$ (or) $\int_{-\infty}^t u(t) dt = r(t)$
4. $\frac{d}{dt} r(t - t_0) = u(t - t_0)$ or $\int_{-\infty}^t u(t - t_0) dt = r(t - t_0)$
5. $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$
6. $x[n]\delta[n - n_0] = x[n_0]\delta[n - n_0]$
7. $u(kt - t_0) = u(t - t_0/k)$, $k \neq 0$.

SYSTEMS AND CLASSIFICATION OF SYSTEMS

A system is a mathematical model of a physical process that relates the input signal to the output signal.

Let x and y be the input and output signals, respectively, of a system. Then, the system is viewed as a transform of x into y .

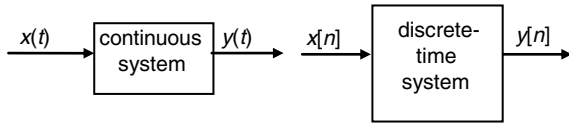
$$y = T\{x\}$$



System with single or multiple input and output signals.

Continuous Time and Discrete-time Systems

If the input and output signals x and y are continuous time (discrete time) signals, then the system is called a continuous time (discrete time) system.



where $x(t)/x[n]$ is the input and $y(t)/y[n]$ is the output. We can also represent them by the notation $x(t) \rightarrow y(t)$ or $x[n] \rightarrow y[n]$.

System with Memory and without Memory

A system is said to be memoryless, if the output at any time depends on only the input at that same time, otherwise the system is said to be system with memory.

Example for memoryless system: $y(t) = kx(t)$, $u[n] = kx[n]$.

The input output relationship of a resistor will be $v(t) = Ri(t)$, where current $i(t)$ is input and $v(t)$ voltage is output. The output at time $t = t_0$ depends on only input at that instant.

A system is said to possess memory if its output signal depends on past or future values of the input signals. The example for continuous time system with memory is capacitor, if input is taken to be the current and the output is the voltage

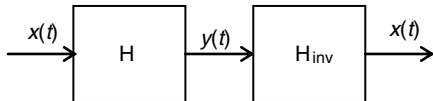
$$V(t) = \frac{1}{C} \int_{-\infty}^t i(T) dT \quad \text{where } C \text{ is capacitance}$$

$y(t) = x^2(t)$ is memoryless system, but $y(t) = x(t^2)$ is system with memory.

Example for system with memory: $y[n] = \sum_{k=-\infty}^n x[k]$, $y[n] = x[n-1]$

Invertible and Non-invertible Systems

A system is said to be invertible either if distinct inputs lead to distinct outputs or if the input of the system can be recovered from the output. When several different inputs result in the same output (as in rectifier), it is impossible to obtain the input from output, and the system is non-invertible.



The system that achieves the inverse operation [of obtaining $x(t)$ from $y(t)$] is the inverse system (H_{inv}) for H . For example, if H is an ideal integrator, then its inverse system (H_{inv}) is an ideal differentiator.

$$x(t) = H_{\text{inv}} \{y(t)\}$$

$$= H_{\text{inv}} H \{x(t)\}$$

$$H_{\text{inv}} H = I, \quad I \text{ is the identity operator.}$$

Cascading a system with its inverse system results in an identity system. The output of a system described by the identity operator is exactly equal to the input. The property

of invertibility is of particular importance in the design of communication systems.

When a transmitted signal propagates through a communication channel, it gets distorted owing to non-ideal frequency response and finite bandwidth of channel. To compensate this distortion, equalizer network will be used, which is the inverse of channel.

Invertible system: $y(t) = x(t - z)$, its inverse system is $x(t) = y(t + z)$.

$$y(t) = \frac{1}{L} \int_{-\infty}^t x(\tau) d\tau; \quad \text{inverse system } x(t) = L \frac{d}{dt} y(t).$$

Non-invertible system: $y(t) = x^2(t)$. The condition for invariability is different inputs and therefore, it should produce different outputs; but for this case, $-x(t)$ and $+x(t)$ both will lead to same output and therefore, non-invertible system.

Causal and Non-causal Systems

If the output of system at present time depends on only the present input $x(t)$ and/or past values of the input, but not on its future values, then the system is causal.

In other words, a casual (also known as physical or non-anticipative) system is one for which the output at any instant t_0 depends only on the value of input $x(t)$ for $t \leq t_0$. In causal system, the output cannot start before the input is applied. A system that violates the conditions of causality is called a non-causal (or anticipative) system. The RC circuit is a causal system, since the capacitor voltage responds only to present and past values of the source voltage. $y(t) = x(-t)$ is not causal because when $t > 0$; consider for $t = +2$, $y(2) = x(-2)$, it seems like a causal system. However, for $t < 0$, consider for $t = -2$, $y(-2) = x(2)$ have output depending on future value, and the causal input-output relation should be valid for all times in a causal system. Therefore, the above-mentioned system is a non-causal system.

$y[n] = x[n] \cos[n+1]$ is a causal system; here, $\cos[n+1]$ is a constant that varies with n , but output $y[n]$ depends on $x[n]$ (present input) only, and therefore, the system is causal.

Example of causal system:

$$y[n] = \sum_{k=-\infty}^n x[k], \quad y(t) = t \cdot x(t), \quad y[n] = x[-n]u[n]$$

Example of non-causal system:

$$y[n] = x[n+1],$$

$$y[n] = \frac{1}{2M+1} \sum_{k=-M}^M x[n-k], \quad y(t) = x(t+2) - x(t-2)$$

All memoryless systems are causal but not vice versa.

Linear Systems and Non-Linear Systems

A system is said to be a linear system, if it satisfies the following two conditions.

1. Additivity: given that $y_1 = T\{x_1\}$,

and $y_2 = T\{x_2\}$, then

$$T\{x_1 + x_2\} = y_1 + y_2, \quad \text{for any signals } x_1 \text{ and } x_2.$$

2. Homogeneity (or scaling): given $T\{x\} = y$, then $T\{ax\} = ay$, for any signal x and y , any scalar a .

The abovementioned conditions can be combined as $T\{a_1x_1 + a_2x_2\} = a_1y_1 + a_2y_2$

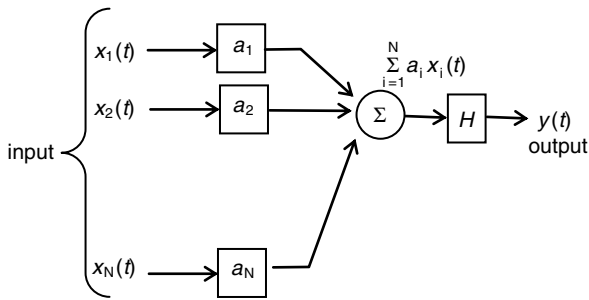
Any system that does not satisfy the above condition is called non-linear system.

Example for non-linear system:

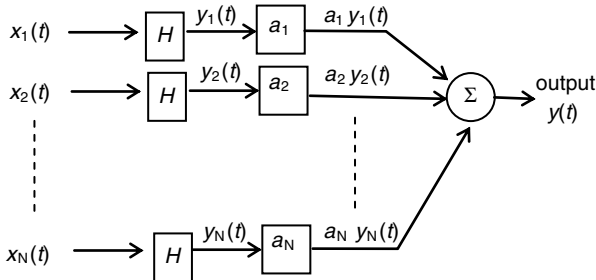
$$y = x^2, y = \sin x.$$

From condition 'B', we can say that a zero input yields zero output for linear systems.

When a system violates either the principle of superposition or the property of homogeneity, the system is said to be 'non-linear'.



The combined operation of amplitude scaling and summation precedes the system operator H for multiple inputs to produce final output $y(t)$.



The system operator H precedes the amplitude scaling for each input; the resulting outputs are summed to produce the overall output $y(t)$.

If these two configurations produce the same output $y(t)$, then the system operator H is linear.

Consider the system output $y(t)$ and input $x(t) = \sum_{i=1}^N a_i x_i(t)$

Then,

$$y(t) = H \left\{ \sum_{i=1}^N a_i x_i(t) \right\} = \sum_{i=1}^N a_i H \{x_i(t)\} = \sum_{i=1}^N a_i y_i(t).$$

Therefore, the system will be linear system.

Consider a system by input–output relation as $y(t) = tx(t)$, if $x(t)$ is expressed as weighted sum $x(t) = \sum_{i=1}^N a_i x_i(t)$

$$\text{Then, } y(t) = t \cdot \sum_{i=1}^N a_i x_i(t) = \sum_{i=1}^N a_i t x_i(t)$$

$y_i(t) = t x_i(t)$ is output due to each input acting individually. Then, $y(t) = \sum_{i=1}^N a_i y_i(t)$ and so the system is linear.

Consider discrete-time system $y[n] = x[n] x[n-1]$

$$\text{Then } y_1[n] = x_1[n] x_1[n-1], y_2[n] = x_2[n] x_2[n-1]$$

If input is $a_1 x_1[n] + a_2 x_2[n]$, then

$$y[n] = (a_1 x_1[n] + a_2 x_2[n]) + (a_1 x_1[n-1] + a_2 x_2[n-1]) \neq a_1 y_1[n] + a_2 y_2[n] \text{ and therefore, the system is non-linear.}$$

Consider system $y(t) = x^2(t)$

$$y_1(t) = x_1^2(t), y_2(t) = x_2^2(t)$$

$$\text{if } x_3(t) = a_1 x_1(t) + a_2 x_2(t) \text{ for linear system } y_3(t) = a_1 y_1(t) + a_2 y_2(t)$$

$$\text{but } y_3(t) = x_3^2(t) = [a_1 x_1(t) + a_2 x_2(t)]^2 \neq a_1 x_1^2(t) + a_2 x_2^2(t) \text{ and therefore, the system is non-linear.}$$

Consider the system

$$\text{If } y[n] = \text{Re} \{x[n]\},$$

Assume $x[n] = p[n] + jq[n]$, then $y[n] = p[n]$ for linearity. (The system must satisfy both additivity and homogeneity properties and that the signals, as well as any scaling constants, are allowed to be complex.)

To check homogeneity property $ay[n] = T\{ax[n]\}$

Here, consider scaling constant $a = j$,

$$x_1[n] = jx[n] = jp[n] - q[n]$$

$$y_1[n] = \text{Re} \{x_1[n]\} = -q[n] \neq jy[n]$$

The system violates homogeneity property, and hence, the system is not linear.

Consider the system $y[n] = 5x[n] + 7$. We can see that the system violates zero-in/zero-out property of linear systems. For zero input $x[n] = 0$, $y[n] = 0$ for linear systems; here, $y[n] = 7$, and therefore, the system is non-linear.

$$\frac{dy(t)}{dt} + 4y(t) = x(t) \text{ is also a linear system and}$$

$$y(t) \frac{dy(t)}{dt} + 4y(t) = x(t) \text{ is non-linear system}$$

Time-invariant and Time-varying Systems

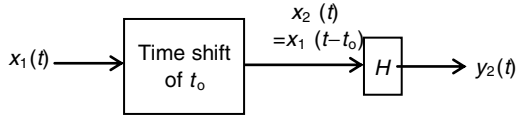
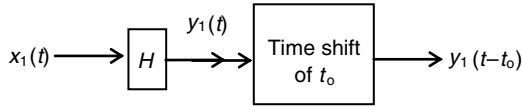
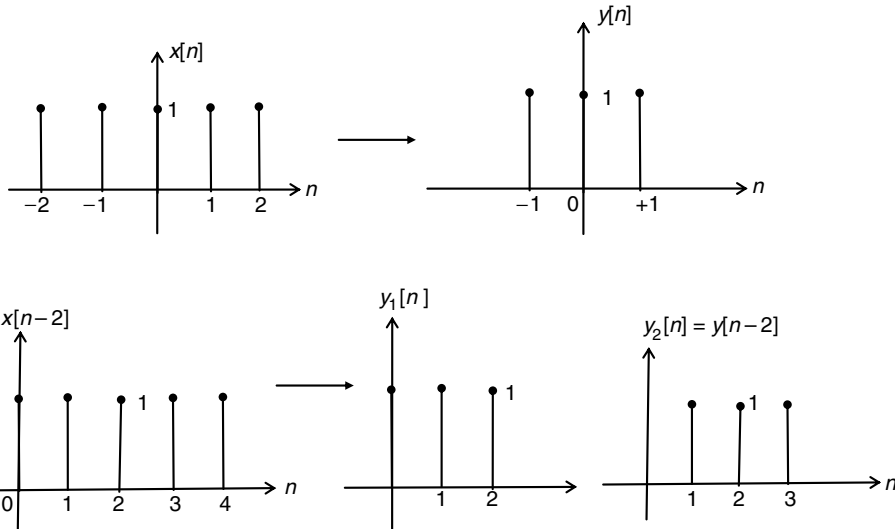
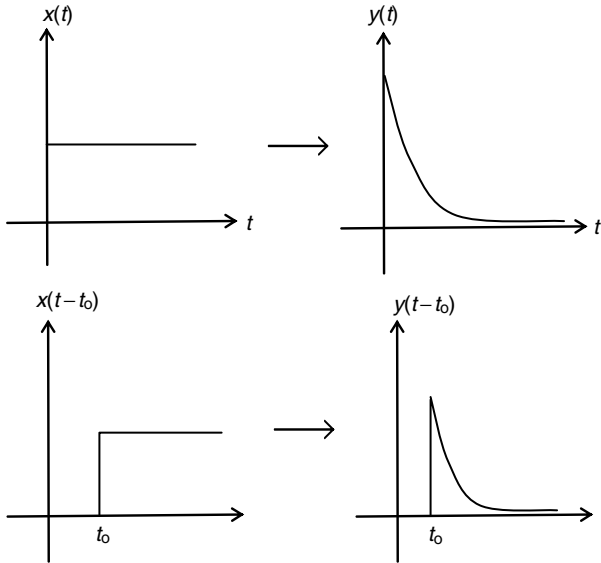
A system is called time invariant if a time shift (delay or advance) in the input signal causes the same time shift in the output signal.

$$T\{x(t - T)\} = y(t - T) \text{ (for C.T)}$$

$$T\{x[n - k]\} = y[n - k] \text{ (for D.T)}$$

A system that does not satisfy above condition is called time-varying system.

This implies that a time-invariant system responds identically no matter when input signal is applied. In other way, the characteristics of a time-invariant system do not change with time.


 Time shift preceding operator H .

 Time shift following operator H . If $y_2(t) = y_1(t - t_0)$, then the system is time invariant.


Stable Systems

A system is bounded input bounded output (BIBO) stable, if for any bounded input x defined by $|x| \leq K_1$. The corresponding output y is also bounded and it is defined by $|y| \leq K_2$, where K_1 and K_2 are finite real constants.

Consider the system

If the system is time invariant, then the delayed output $y(t - t_0)$ can also be obtained by first delaying the input $x(t)$ before applying it to the system.

Consider a system specified by $y(t) = e^{-2t}x(t - 1)$ for the delayed input $x(t - t_0)$, and the output is

$$y_1(t) = e^{-2t}x(t - t_0 - 1); \text{ however, delayed output}$$

$y_2(t) = y(t - t_0) = e^{-2(t-t_0)}x(t - t_0 - 1) \neq y_1(t)$, and therefore, the system is time varying.

Consider another system $y(t) = (\sin t)x(t + 1)$ for delayed input $x(t - t_0)$, output $y_1(t) = \sin t$.

$$x(t - t_0 + 2) \text{ the delayed output.}$$

$y_2(t) = y(t - t_0) = \sin(t - t_0)x(t - t_0 + 2) \neq y_1(t)$ and therefore, the system is time varying.

Consider a discrete-time system

$$y[n] = \sin(x[n]). \text{ For delayed input } x[n - n_0],$$

$$\text{output } y_1[n] = \sin(x[n - n_0])$$

When output is delayed,

$$y_2[n] = y[n - n_0] = \sin(x[n - n_0]).$$

Here,

$$y_1[n] = y_2[n], \text{ and therefore, the system is time invariant.}$$

Consider a discrete-time system, $y[n] = x[2n]$;

for delayed input, $x[n - n_0]$, the output $y_1[n] = x[2n - n_0]$.

When output is delayed, $y_2[n] = y[n - n_0] = x[2(n - n_0)]$ $y_1[n] \neq y_2[n]$ and therefore, the system is time variant.

Therefore, the above system is time variant, $y_2[n] \neq y_1[n]$.

If the system is linear and also time invariant, then it is called a linear time-invariant (LTI) system.

$$y[n] = \frac{1}{3} (x[n] + x[n - 2] + x[n - 3])$$

If input is finite value,

$$|x[n]| \leq k < \infty \text{ for all } n,$$

$$\text{Then, } |y[n]| = \left| \frac{1}{3} (x[n] + x[n - 2] + x[n - 3]) \right|$$

$$\leq \frac{1}{3}(|x[n]| + |x[n-2]| + |x[n-3]|)$$

$$\leq \frac{1}{3}(k_1 + k_1 + k_1) = k_1 < \infty$$

The output is also finite for finite input, and therefore, system is BIBO stable.

Consider a continuous time system

$y(t) = a'x(t)$. If input is finite value $|x(t)| \leq k_1 < \infty$ for all n .

Then, $|y(t)| = |a'x(t)| = |a'| |x(t)|$

With $a > 1$, the multiplying factor a' diverges for increasing t , and therefore, system is unstable for $a > 1$; for $a < 1$, the system will be stable.

Consider the continuous time system $y(t) = tx(t)$. Even for bounded input $|x(t)| \leq k < \infty$, the output is unbounded $|y(t)| = |t| |x(t)| = |t| k$.

Therefore, the system is unstable.

Consider the continuous time system $y(t) = e^{2x(t)}$

When $x(t)$ is finite

$|x(t)| < k < \infty$

$-k < x(t) < k$, then $y(t) = e^{2x(t)}$, that is,

$e^{-2k} < |y(t)| < e^{2k} y(t)$ is also finite. Therefore, the system is stable.

LINEAR TIME-INVARIANT SYSTEMS

Impulse Response

The impulse response $h(t)$ of a continuous time LTI system (represented by T) is defined to be the response of the system, when the input is $\delta(t)$, that is, $h(t) = T\{\delta(t)\}$.

In continuous time case, a true impulse signal having zero width and infinite amplitude cannot physically be generated and is usually approximated by a pulse of large amplitude and brief duration. Thus, impulse response may be interpreted as the system behaviour in response to a high energy input of extremely brief duration.

Response to an Arbitrary Input $x(t)$

Given the impulse response $h(t)$, we determine the output $y(t)$ due to an arbitrary input signal $x(t)$ by expressing the input as a weighted super position of time shifted impulses. By linearity and time invariance, the output signal must be a weighted super position of time-shifted impulse responses. This weighted super position is termed as the convolution integral for continuous time system, and the convolution sum for discrete-time system.

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

input $x(t)$ can be expressed as

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$$

Output $y(t) = T\{x(t)\}$

$$= T\left\{\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau\right\} = \int_{-\infty}^{\infty} x(\tau) T\{\delta(t-\tau)\}d\tau$$

Since the system is time invariant

$T\{\delta(t-\tau)\} = h(t-\tau)$ ($\because$ impulse response)

$$\text{Therefore, } y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

Convolution Integral

The convolution of two signals $x(t)$, $h(t)$ is denoted by $y(t) = x(t) * h(t)$

$$= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

The output of any LTI system is the convolution of the input and the impulse response of the system



Figure 3 Continuous time LTI system

Properties of Convolution Integral

1. Commutative $x(t) * h(t) = h(t) * x(t)$
2. Associative $\{x(t) * h_1(t)\} * h_2(t) = x(t) * \{h_1(t) * h_2(t)\}$
3. Distributive $x(t) * \{h_1(t) + h_2(t)\} = x(t) * h_1(t) + x(t) * h_2(t)$
4. $y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$
5. $x_1(t) * x_2(t) = c(t)$, $x_1(t-T_1) * x_2(t-T_2) = c(t-T_1-T_2)$
6. Convolution with an impulse $x(t) * \delta(t) = x(t)$, $x(t-t_1) * \delta(t-t_2) = x(t-t_1-t_2)$
7. The width property: if $x_1(t)$ is defined for T_1 duration and $x_2(t)$ is defined for T_2 duration, then duration (width) of $x_1(t) * x_2(t)$ is $T_1 + T_2$.
8. Causality: if $h(t)$ and $x(t)$ are causal systems, then

$$y(t) = x(t) * h(t) = \int_{0-}^t x(\tau)h(t-\tau)d\tau, t \geq 0$$

$$= 0, t < 0$$

$$(\text{or}) y(t) = \int_{0-}^t h(\tau)x(t-\tau)d\tau, t \geq 0$$

$$= 0, t < 0$$

Step Response

The step response $s(t)$ of a continuous time LTI system is defined to be the response of the system when input is $u(t)$; unit step response $s(t) = T\{u(t)\}$

$s(t)$ can be easily determinate by

$$s(t) = h(t) * u(t)$$

$$= \int_{-\infty}^{\infty} h(\tau)u(t-\tau)d\tau = \int_{-\infty}^t h(\tau)d\tau = \int_0^{\infty} h(t-\tau)d\tau$$

$$\text{or } h(t) = \frac{d}{dt} s(t).$$

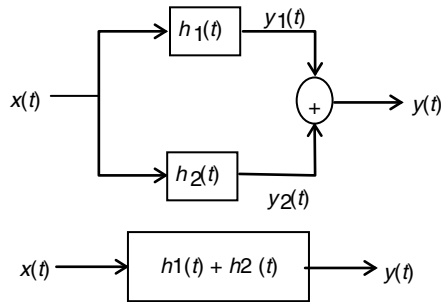
Convolution Table

S. No.	$x_1(t)$	$x_2(t)$	$x_1(t) * x_2(t) = x_2(t) * x_1(t)$
1	$x(t)$	$\delta(t - T)$	$x(t - T)$
2	$u(t)$	$u(t)$	$tu(t)$
3	$e^{at} u(t)$	$e^{bt} u(t)$	$\frac{e^{at} - e^{bt}}{a - b} u(t), a \neq b$
4	$e^{at} u(t)$	$e^{at} u(t)$	$te^{at} u(t)$
5	$e^{at} u(-t)$	$e^{bt} u(-t)$	$\frac{e^{at} - e^{bt}}{b - a} u(-t)$

Interconnected Systems

A large, more complex system can often be viewed as the interconnection of several smaller subsystems, each of which is easier to characterize. By understanding the characterizations of these subsystems, it becomes simpler to analyse such large systems.

Consider two continuous time LTI systems that are in parallel,

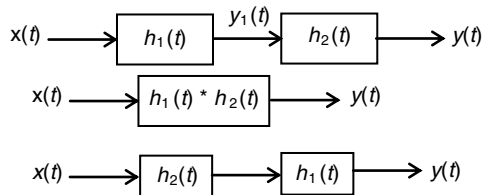


$$y_1(t) = h_1(t) * x(t), y_2(t) = h_2(t) * x(t)$$

$$\begin{aligned} y(t) &= y_1(t) + y_2(t) = h_1(t) * x(t) + h_2(t) * x(t) \\ &= [h_1(t) + h_2(t)] * x(t) \end{aligned}$$

This is another interpretation of distributive property of convolution.

Consider two continuous time LTI systems in cascade



$$y_1(t) = x(t) * h_1(t)$$

$$\begin{aligned} y(t) &= y_1(t) * h_2(t) \\ &= x(t) * h_1(t) * h_2(t) \\ &= x(t) * \{h_1(t) * h_2(t)\} \end{aligned}$$

This is another interpretation of associate property and commutative property of convolution.

Properties of Continuous Time LTI Systems

Systems With or Without Memory

A system is memoryless, if its output at any time depends only on the value of the input at that same time.

$$\text{output } y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau$$

If $y(t)$ needs to be dependent only present input $x(t)$, then $h(\tau) x(t - \tau) = kx(t)$; this is possible when $h(\tau)$ is defined for only $\tau = 0$,

that is, $h(t) = 0$ for $t \neq 0$.

The corresponding Impulse response $h(t)$ of memoryless system is simple $h(t) = k\delta(t)$. Therefore, if $h(t_0) \neq 0$ for $t_0 \neq 0$, and the continuous time LTI system has memory.

Causality

The output of a causal system depends only on the present and past values of input to the system.

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau$$

$y(t)$ should not depend on future values, that is, $h(t) = 0$ for $t < 0$

For a causal continuous time LTI system, we have $h(t) = 0, t < 0$.

The output of causal LTI system is

$$y(t) = \int_0^t h(\tau) x(t - \tau) d\tau = \int_0^t x(\tau) h(t - \tau) d\tau$$

Stability

A system is bounded input–bounded output (BIBO) stable if the output is guaranteed to be bounded for every bounded input.

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau$$

$$|y(t)| < \infty, \text{ for } |x(t)| < \infty \text{ then}$$

$$\begin{aligned} |y(t)| &= \left| \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \right| \\ &= \int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty \text{ as } |x(t)| < \infty. \end{aligned}$$

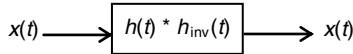
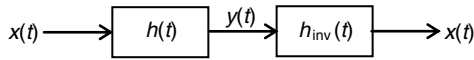
A continuous time LTI system is BIBO stable, if its impulse response is absolutely integrable

$$\text{(i.e., } \int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty)$$

Invertibility

If an LTI system is invertible, then it has an LTI inverse system. When the inverse system is connected in series with

original system, it produces an output equal to the input to the first system.



$$x(t) = x(t) * \{h(t) * h_{\text{inv}}(t)\}$$

This will be valid only when

$$h(t) * h_{\text{inv}}(t) = \delta(t).$$

Discrete-time LTI Systems

1. Impulse response $h[n] = T\{\delta[n]\}$

2. Response to an arbitrary input $x[n]$ is

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

3. Convolution $y[n] = x[n] * h[n]$

$$= \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

4. Properties of convolution sum

$$x[n] * h[n] = h[n] * x[n] - \text{Commutative}$$

– Associative

$$x[n] * \{h_1[n] + h_2[n]\} = x[n] * h_1[n] + x[n] * h_2[n]$$

– Distributive

5. Shifting property $x_1[n] * x_2[n] = c[n]$

$$\text{Then, } x_1[n - n_1] * x_2[n - n_2] = c[n - n_1 - n_2]$$

6. Convolution with impulse $x[n] * \delta[n] = x[n]$,

$$x[n - n_1] * \delta[n - n_2] = x[n - n_1 - n_2]$$

7. The width property, if $x_1[n]$ and $x_2[n]$ have finite widths of w_1 and w_2 , respectively, then the width of $x_1[n] * x_2[n]$ is $w_1 + w_2$. The width of a signal is one less than the number of elements.

8. Causal system output; if $x[n]$ and $h[n]$ are causal, then

$$y[n] = \sum_{k=0}^{\infty} x[k]h[n-k]$$

9. Convolution sum table:

S.No.	$x_1[n]$	$x_2[n]$	$x_1[n] * x_2[n]$
1	$\delta[n - k]$	$x[n]$	$x[n - k]$
2	$u[n]$	$u[n]$	$(n + 1) \cdot u[n]$
3	$a^n u[n]$	$b^n u[n]$	$\left[\frac{a^{n+1} - b^{n+1}}{a - b} \right] u[n], a \neq b$
4	$u[n]$	$nu[n]$	$\frac{n(n+1)}{2} u[n]$
5	$a^n u[n]$	$a^n u[n]$	$(n + 1)a^n u[n]$

10. When two LTI discrete-time systems with impulse response $h_1[n]$ and $h_2[n]$, respectively, are connected in parallel, the impulse response of the parallel

composite system is $h_1[n] + h_2[n]$. If these systems are connected in cascade (in any order), the impulse response of the composite system is $h_1[n] * h_2[n] = h_2[n] * h_1[n]$.

11. Step response

$$s[n] = h[n] * u[n] = \sum_{k=0}^{\infty} h[n-k] = \sum_{k=0}^{\infty} h[k]$$

$$h[n] = s[n] - s[n-1]$$

12. If $h[n] \neq 0$, for $n \neq 0$, the discrete-time LTI system has memory (or) $h[n] = 0$, for $n \neq 0$ system is memoryless.

13. Causality condition is $h[n] = 0$ for $n < 0$, causal system output.

$$y[n] = \sum_{k=0}^{\infty} h[k]x[n-k] = \sum_{k=0}^{\infty} x[k]h[n-k]$$

14. A DT, LTI system is BIBO stable if $\sum_{k=-\infty}^{\infty} |h[k]| < \infty$.

15. Invertibility: $h[n] * h_{\text{inv}}[n] = \delta[n]$

The steps involved in calculation of convolution.

Continuous time convolution	Discrete-time convolution
$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$ $= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$ $= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$	$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$ $= \sum_{k=-\infty}^{\infty} x[k]h[n-k]$ $= \sum_{k=-\infty}^{\infty} h[k]x[n-k]$
1 $x(t) \rightarrow x(\tau)$ (or) $h(t) \rightarrow h(\tau)$	1 $h[n] \rightarrow h[k]$, (or) $x[n] \rightarrow x[k]$
2 Folding (or) flipping $x(-\tau)$ or $h(-\tau)$	2 Folding (or) flipping $x[-k]$ or $h[-k]$
3 Shifting $x(t - \tau)$ or $h(t - \tau)$	3 Shifting $x[n-k]$ (or) $h[n-k]$
4 Multiplication $x[t - \tau]h[\tau]$ (or) $h[t - \tau]x[\tau]$	4 Multiplication $h[k]x[n-k]$ (or) $x[k]h[n-k]$
5 Integration	5 Summation

Solved Examples

Example 1

Consider the following signals $x(t) = \sin \pi t + \cos 2\pi t$.

$$y(t) = \sin 2t + \sin 5t.$$

$$z(t) = \sin t + \cos \pi t$$

Periodic signals are

- (A) $x(t)$ and $z(t)$ (B) $x(t)$ and $y(t)$
 (C) $y(t)$ and $z(t)$ (D) Only $y(t)$

Solution

$x(t)$ is periodic with period $T = 2$ s.

$y(t)$ is periodic with period $T = 2\pi$ s.

For $z(t)$, fundamental time period cannot be determined.

Example 2

Consider the following signals with $z > 0$.

- (a) $e^{-zt} u(t)$ (b) $e^{zt} u(t)$ (c) $tu(t)$

The energy signals are

- (A) a and b (B) b only (C) a only (D) all

Solution

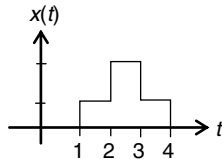
All integrable signals are energy signals

'a' is only integrable signal

'b', 'c' are not integrable signals.

Example 3

Consider the voltage waveform $x(t)$

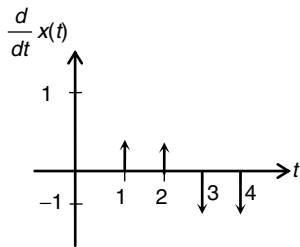


The equation for $x(t)$ is

- (A) $u(t-1) - u(t-2) - u(t-3) - u(t-4)$
 (B) $u(t-1) + u(t-2) - u(t-3) - u(t-4)$
 (C) $u(t-1) + u(t-2) + u(t-3) - u(t-4)$
 (D) $u(t-1) - u(t-2) + u(t-3) + u(t-4)$

Solution

Differentiate $x(t)$

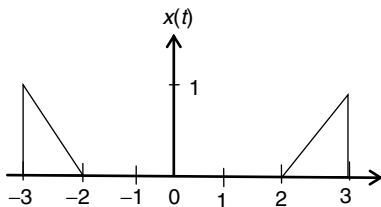


$\frac{d}{dt} x(t) = \delta(t-1) + \delta(t-2) - \delta(t-3) - \delta(t-4)$ After integration,

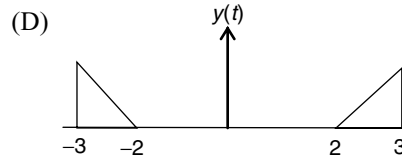
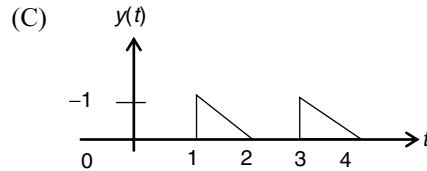
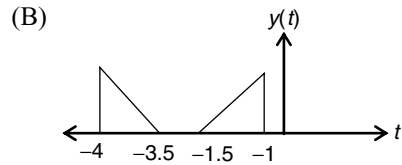
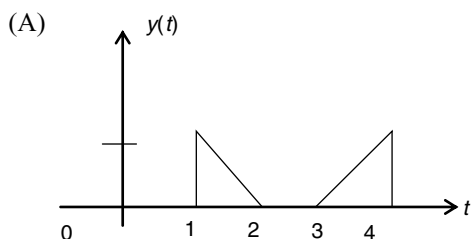
$$x(t) = u(t-1) + u(t-2) - u(t-3) - u(t-4)$$

Example 4

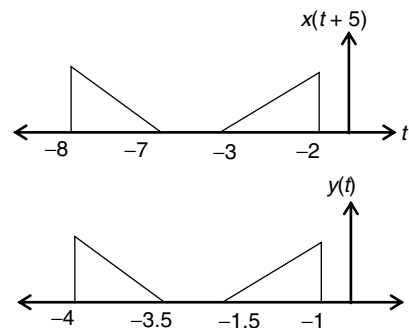
The signal $x(t)$ is depicted as



The signal $y(t)$ is related to $x(t)$ as $y(t) = x(2t+5)$, the sketch of $y(t)$ is



Solution



Example 5

The even part of a function

$$x(n) = u(n) + u(-n)$$

- (A) $u(n) - u(-n)$ (B) $u(n) + u(-n)$
 (C) $u(-n) - u(n)$ (D) $2u(n)$

Solution

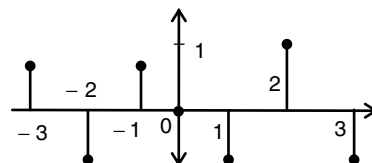
$$\text{Even part of } x(n) = \frac{x(n) + x(-n)}{2}$$

$$= \frac{u(n) + u(-n) + u(-n) + u(n)}{2}$$

$$x_1(n) = u(n) + u(-n)$$

Example 6

$x(n)$ is given as



Choose the correct option for

$$y(n) = x(1-n) - x(n-1)$$

- (A) $\{-2, 2, -2, 0, 2, -2, 2\}$ (B) $\{-1, 1, -1, 0, 2, 2, 2\}$
 (C) $\{2, -2, 2, 0, 2, -2, 2\}$ (D) $\{-2, 2, -2, 0, -2, 2, -2\}$

Solution

$$x(n-1) = \left\{ 1, -1, \underset{\uparrow}{1}, 0, -1, 1, -1 \right\}$$

$$x(1-n) = \left\{ -1, 1, \underset{\uparrow}{-1}, 0, 1, -1, 1 \right\}$$

$$x(1-n) - x(n-1) = \left\{ -2, 2, \underset{\uparrow}{-2}, 0, 2, -2, 2 \right\}$$

Example 7

A discrete-time signal

$$x(n) = \cos\left(\frac{\pi}{7}n\right) + \sin\left(\frac{\pi n}{8} + \frac{1}{2}\right), \text{ the period of } x(n)$$

- (A) Periodic with period 112
 (B) Periodic with period 223
 (C) Periodic with period 334
 (D) Not periodic

Solution

$$N_1 = 14, N_2 = 16$$

$$N = \text{L.C.M}(14, 16) = 112$$

Therefore, $x(n)$ is periodic with period 112.

Example 8

$$x(n) = \left\{ 1, 2, \underset{\uparrow}{-3}, 2, 1 \right\}, h(n) = \left\{ 4, \underset{\uparrow}{-2}, 1 \right\}$$

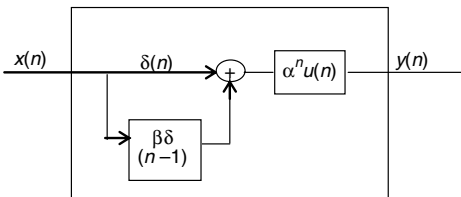
Find $x[n]*h(n)$.

- (A) $\left\{ 4, 6, -15, \underset{\uparrow}{16}, -3, 0, 1 \right\}$
 (B) $\left\{ \underset{\uparrow}{4}, 6, -15, 16, -3, 0, 1 \right\}$
 (C) $\left\{ 4, 6, 15, \underset{\uparrow}{-16}, 3, 0, 1 \right\}$
 (D) $\left\{ \underset{\uparrow}{4}, 6, 15, -16, 3, 0, 1 \right\}$

Solution

$$x(n) * h(n) = h(n+2) + 2h(n+1) - 3h(n) + 2h(n-1) + h(n-2)$$

4	-2	1				
	8	-4	2			
		-12	6	-3		
			8	-4	2	
				4	-2	1
4	6	-15	16	-3	0	1

Example 9

What is the transfer function $h(n)$?

$$(A) \alpha^n + 1 u(n) + \beta \alpha^{n-1} u(n-1) = h(n)$$

$$(B) \alpha^n u(n-1) + \beta \alpha^{n-1} u(n) = h(n)$$

$$(C) \alpha^n u(n) + \beta \alpha^n u(n-1) = h(n)$$

$$(D) \alpha^n u(n) + \beta \alpha^{n-1} u(n-1) = h(n)$$

Solution

Blocks are parallel. Additional blocks are cascaded convolution, and therefore,

$$\frac{x(n)}{\delta(n) + \beta \delta(n-1)} \longrightarrow \frac{\alpha^n u(n)}{\alpha^n u(n)} y(n)$$

$$\frac{x(n)}{[\delta(n) + \beta \delta(n-1)] * \alpha^n u(n)} y(n)$$

$$\frac{x(n)}{[\delta(n) * \alpha^n u(n) + \beta \delta(n-1) * \alpha^n u(n)]} y(n)$$

From basic formulae

$$x(t) \delta(t - t_0) = x(t - t_0)$$

$$x(t) \delta(t - t_0) = x(t_0) \delta(t - t_0)$$

$$\int_{-\infty}^{\infty} x(t) * \delta(t - t_0) dt = x(t_0)$$

$$\frac{x(n)}{\alpha^n u(n) + \beta \alpha^{n-1} u(n-1)} y(n)$$

Example 10

The result of $u(t) * h(t)$, when $h(t) = \begin{cases} e^{at}, & t < 0 \\ e^{-bt}, & t > 0 \end{cases}$

$$(A) \frac{1}{a} e^{at} + \frac{1}{a} (1 - e^{at}) u(t) + \frac{1}{b} (1 - e^{-bt}) u(t)$$

$$(B) \frac{1}{a} e^{at} - \frac{1}{a} (1 - e^{at}) u(t) + \frac{1}{b} (1 - e^{-bt}) u(t)$$

$$(C) \frac{1}{a} e^{at} - \frac{1}{a} (1 - e^{at}) u(t) + \frac{1}{b} (1 - e^{-bt}) u(t)$$

$$(D) \frac{1}{a} e^{-at} + \frac{1}{a} (1 - e^{-at}) u(t) + \frac{1}{b} (1 - e^{-bt}) u(t)$$

Solution

$$\text{For } t < 0, \int_{-\infty}^0 e^{at} dt = \frac{e^{at}}{a}$$

$$\text{For } t \geq 0, \int_{-\infty}^0 e^{at} dt + \int_0^t e^{-bt} dt = \frac{1}{a} + \frac{1}{b} - \frac{1}{b} e^{-bt} =$$

$$\frac{1}{a} + \frac{1}{b} (1 - e^{-bt})$$

$$= u(t) * h(t) = \frac{e^{at}}{a} (1 - u(t)) + \left(\frac{1}{a} + \frac{1}{b} (1 - e^{-bt}) \right) u(t)$$

$$= \frac{1}{a} e^{at} + \frac{1}{a} (1 - e^{at}) u(t) + \frac{1}{b} (1 - e^{-bt}) u(t)$$

Example 11

The discrete-time input $x(n)$ and output $y(n)$ relationship as $y(n) = \sqrt{n} x(n)$, then properties of the system are

- (A) linear, time invariant, causal, stable
 (B) non-linear, time invariant, causal, stable
 (C) non-linear, time variant, causal, unstable
 (D) Linear, time invariant, causal, unstable

Solution

$$y_1(n) \rightarrow \sqrt{nx_1(n)}$$

$$y_2(n) \rightarrow \sqrt{nx_2(n)}$$

$$y_1(n) + y_2(n) \rightarrow \sqrt{n(x_1(n) + x_2(n))} \neq \sqrt{nx_1(n)} + \sqrt{nx_2(n)}$$

It is not additive, and therefore, it is non-linear

$$y(n) \rightarrow \sqrt{nx(n)}$$

$$y(n - n_0) \rightarrow \sqrt{(n - n_0)x(n - n_0)} \neq \sqrt{nx(n - n_0)}$$

Therefore, it is time variant.

At any discrete time, $n = n_0$. The response depends only on the excitation at that time. Therefore, 'causal' if the excitation is a constant the response is unbounded as ' n ' approaches infinity so unstable.

Example 12

A continuous time linear system with input $x(t)$ and output $y(t)$ yields the following input and output pairs:

$$x(t) = e^{-jt} \leftrightarrow y(t) = e^{j3t}$$

$$x(t) = e^{jt} \leftrightarrow y(t) = e^{-j3t}$$

If $x(t) = \sin(t - 1)$. The corresponding $y(t)$ is

- (A) $3 \sin(t - 1)$ (B) $\sin(3t - 1)$
 (C) $\sin(3t - 3)$ (D) $\sin(t - 3)$

Solution

$$\sin(t - 1) = \frac{e^{j(t-1)} - e^{-j(t-1)}}{2j} = \frac{e^{-j}e^{jt} - e^je^{-jt}}{2j}$$

Since system is linear

$$\begin{aligned} y(t) &= \frac{e^{-j}e^{j3t} - e^je^{-j3t}}{2j} \\ &= \frac{e^{j(3t-1)} - e^{-j(3t-1)}}{2j} = \sin(3t - 1) \end{aligned}$$

Example 13

The rise of cosine pulse $x(t)$ is defined as

$$x(t) = \begin{cases} 1 + \cos \omega t, & -\frac{\pi}{\omega} \leq t \leq \frac{\pi}{\omega} \\ 0, & \text{otherwise} \end{cases}$$

The total energy of $x(t)$ is

- (A) $\frac{\pi}{\omega}$ (B) $\frac{2\pi}{\omega}$ (C) $\frac{3\pi}{\omega}$ (D) $\frac{4\pi}{\omega}$

Solution

$$\text{Energy} = \int_{-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} (1 + \cos \omega t)^2 dt$$

$$= 2 \int_0^{\frac{\pi}{\omega}} \left(\frac{1}{2} \cos 2\omega t + 2 \cos \omega t + \frac{3}{2} \right) dt = 2 \left(\frac{3}{2} \right) \left(\frac{\pi}{\omega} \right) = \frac{3\pi}{\omega}$$

Example 14

A signal $v(n)$ is defined by

$$v(n) = \begin{cases} 1; & n = 1 \\ -1; & n = -1 \\ 0; & n = 0 \text{ and } |n| > 1 \end{cases}$$

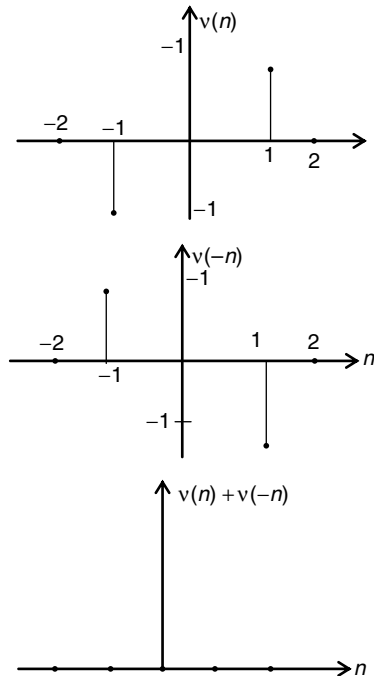
Which is the value of the composite signal defined as $v[n] + v[-n]$

- (A) 0 for all integer values of ' n '
 (B) '2' for all integer value of ' n '
 (C) 1 for all integer values of ' n '
 (D) -1 for all integer values of ' n '.

Solution

$$v(n) + v(-n) = 0$$

For all integer values of n

**Example 15**

The impulse response of a L.T.I system is given as $h(n)$

$$= \left(\frac{-1}{4} \right)^n u(n), \text{ the step response is}$$

- (A) $\frac{1}{4} \left[5 + \left(\frac{-1}{5} \right)^n \right]$ (B) $\frac{1}{5} \left[4 + \left(\frac{-1}{4} \right)^n \right]$
 (C) $\frac{1}{4} \left[5 - \left(\frac{-1}{4} \right)^n \right]$ (D) $\frac{1}{5} \left[4 - \left(\frac{-1}{5} \right)^n \right]$

SolutionFor $n < 0$, $s(n) = 0$

$$\text{For } n \geq 0, s(n) = \sum_{k=0}^n \left(\frac{-1}{4}\right)^k$$

$$= \frac{1 - \left(\frac{-1}{4}\right)^{n+1}}{1 + \frac{1}{4}}$$

$$= \frac{1}{5} \left(4 + \left(\frac{-1}{4}\right)^n \right)$$

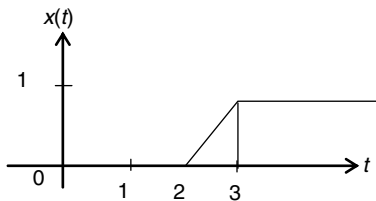
EXERCISES**Practice Problems 1**

Direction for questions 1 to 30: Select the correct alternative from the given choices.

1. The period of signal $x(t) = \sin t + \cos \sqrt{2} t$ is

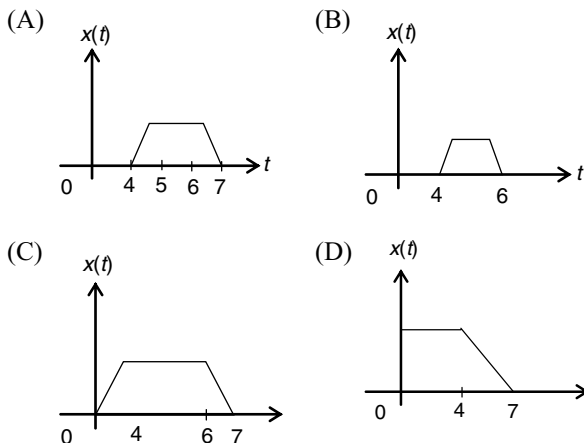
- (A) $\pi/\sqrt{2}$
 (B) 2π
 (C) 3π
 (D) not periodic

2. Consider the following signal $x(t)$

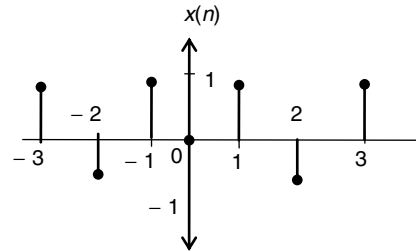


The function that describe the $x(t)$ is

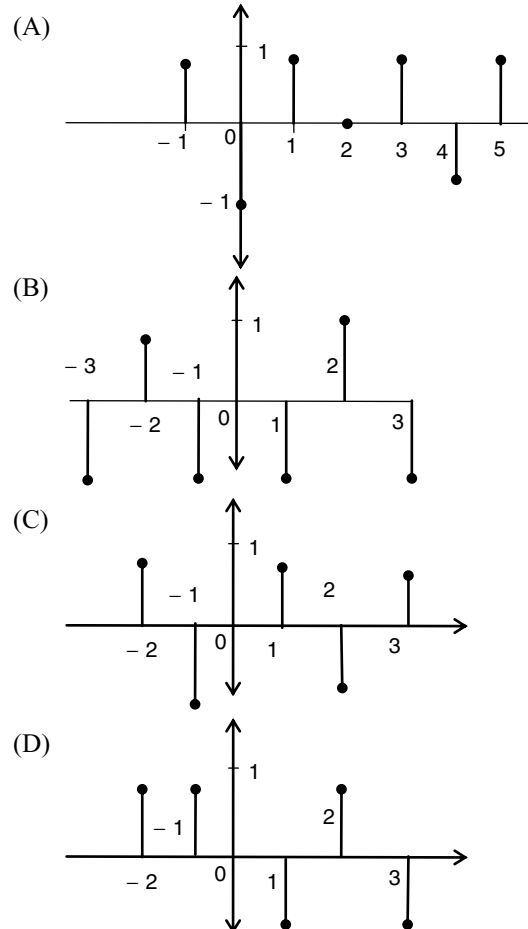
- (A) $(t-2)u(t-2) + (t-3)u(t-3)$
 (B) $(t-2)u(t-2) - (t-3)u(t-3)$
 (C) $(t-2)u(t-2) + (t+3)u(t+3)$
 (D) $(t-2)u(t-2) - (t+3)u(t+3)$
3. A signal is described by $x(t) = r(t-4) - r(t-5) - r(t-6) + r(t-7)$, where $r(t)$ is a unit ramp function starting at $t = 0$. The signal $x(t)$ is represented as



4. $x(n)$ is given as



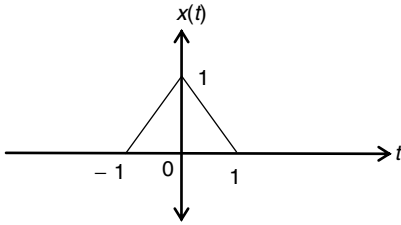
Choose the sketch for the signal $x(2-n)$?



5. A discrete-time signal $x(n]$ defined as $x(n) = e^{-j\left(\frac{n}{4} + \pi\right)}$ the period of signal $x[n]$ is?

- (A) periodic with $N = 8$ (B) periodic with $N = 16$
 (C) periodic with $N = 4$ (D) None of the above

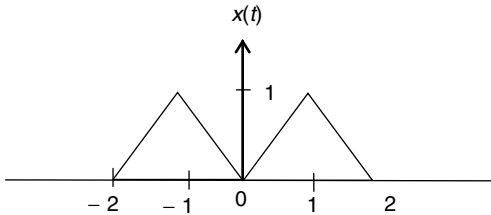
6. The signal $x(t)$ is



$x(t)$ is applied to a differentiator defined by $y(t) = \frac{dx(t)}{dt}$. Find the total energy of $y(t)$.

- (A) 2 (B) 1 (C) 3 (D) 4

7. The signal $x(t)$ is



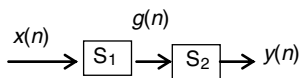
The total energy of $x(t)$ is

- (A) 4/3 (B) 3/4 (C) 1 (D) 5/4

8. The sinusoidal signal $x(t) = 3 \cos(100t + \pi/4)$ is passed through a square law device defined by the input and output relation $y(t) = x^2(t)$. The DC component in the signal is

- (A) 9 (B) 9/2 (C) 9/4 (D) 9/8

9. Consider the following two systems S_1 and S_2 as shown



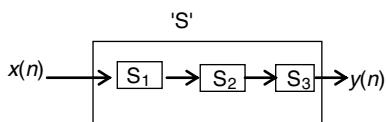
S_1 : causal L.T.I $g(n) = g(n-1) + x(n)$

S_2 : causal L.T.I $y(n) = ay(n-1) + bg(n)$

The difference equation for cascaded system is $y(n) = -\frac{1}{4}y(n-2) + \frac{5}{4}y(n-1) + \frac{1}{2}x(n)$ the values of a and b are

- (A) $\frac{1}{4}, \frac{1}{2}$ (B) $\frac{1}{2}, \frac{1}{2}$ (C) $\frac{1}{2}, \frac{1}{4}$ (D) $\frac{1}{4}, \frac{1}{4}$

10. Three discrete-time systems S_1 , S_2 , and S_3 are connected in cascade to form a new system 'S' as



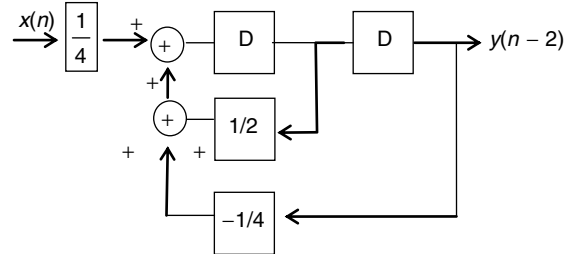
Consider the following statements.

- (1) If S_1 , S_2 , and S_3 are linear, then 's' is linear.
 (2) If S_1 , S_2 , and S_3 are non-linear, then 's' is non-linear.
 (3) If S_1 , S_2 , and S_3 are causal, then 's' is causal.
 (4) If S_1 , S_2 , and S_3 are time invariant, then 's' is the time invariant.

True statements are

- (A) 1, 2, 3 (B) 2, 3, 4 (C) 1, 3, 4 (D) All

11. The system shown as



- (A) stable and causal
 (B) stable but not causal
 (C) causal but unstable
 (D) unstable but not causal

12. The following signal

$$y(n) = \sum_{k=-n-n_0}^{n+n_0} x[k] \text{ is}$$

- (A) Causal (B) Non-causal
 (C) Both (D) Anti-causal

13. $x(n) = \{1, 2, 0, 2, 1\}$, $h(n) = x(n)$. Find $x(n) * h(n)$.

- (A) $\{1, 4, 4, 4, 10, 4, 4, 4, 1\}$
 (B) $\{1, 4, 4, 4, 10, 4, 4, 4, 1\}$
 (C) $\{1, 2, 3, 4, 10, 4, 3, 2, 1\}$
 (D) $\{1, 2, 3, 4, 10, 4, 3, 2, 1\}$

14. A discrete-time LTI system with input $u[n]$ produces output $\tilde{x}(n)$, then output due to the input $nu(n)$

- (A) $u(n)$ (B) $u(n-1)$
 (C) $u(n) - u(n-1)$ (D) $u(n) + u(n-1)$

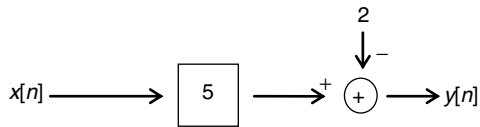
15. A discrete-time system has impulse response $h(n) = a^n u(n+2)$ $|a| < 1$, which one of the following statement is correct? The system is

- (A) stable, causal and memoryless
 (B) stable, non-causal and has memory
 (C) stable, non-causal and memoryless
 (D) unstable, non-causal and memoryless

16. Which one of the following systems having an impulse response given is stable and causal LTI system?

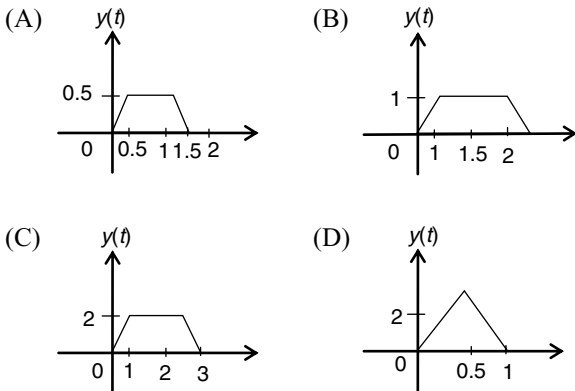
- (A) $h_1(t) = 1$ (B) $h_2(t) = u(t)$
 (C) $h_3(t) = \frac{u(t+1)}{t+1}$ (D) $h_4(t) = e^{-2t} u(t)$

17. The discrete-time input $x[n]$ and output $y[n]$ relationship as



Determine system properties.

- (A) non-linear, time invariant, causal, stable
 (B) non-linear, time variant, causal, unstable
 (C) linear, time invariant, causal, stable
 (D) non-linear, time invariant, non-causal, stable
18. Suppose that $x(t) = \begin{cases} 1 & 0 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}$ and $h(t) = x(2t)$, then $y(t) = x(t) * h(t)$ is



19. Consider the following systems $y_1(t) = x(t+4)$, $y_2(t) = \int_{-\infty}^t x(T) dT$, $y_3(t) = 2 \frac{d}{dt} x(t)$

The non-invertible system is

- (A) $y_1(t)$ and $y_2(t)$ (B) $y_1(t)$ and $y_3(t)$
 (C) Only $y_3(t)$ (D) All the above
20. If the signals $f_1(t)$ and $f_2(t)$ are orthogonal over the interval (t_1, t_2) , then

- (A) $\int_{t_1}^{t_2} f_1(t) f_2(t) dt = \frac{\pi}{2}$ (B) $\int_{t_1}^{t_2} f_1(t) f_2(t) dt = 0$
 (C) $\int_{t_2}^{t_1} f_1(t) * f_2(t) dt = 0$ (D) $\int_{t_1}^{t_2} f_1(t) f_2(t) dt = 1$

21. What are the even and odd components of the signal $f(t) = e^{jt}$?

- (A) $\cos t, j \sin t$ (B) $\cos t, -j \sin t$
 (C) $\sin t, j \cos t$ (D) $-\cos t, j \sin t$

22. A continuous time LTI system has 'Step response' $s(t)$ and impulse response $h(t)$ are related by

- (A) $h(t) = \int s(t) dt$ (B) $h(t) = \frac{ds(t)}{dt}$
 (C) $h(t) = s(t) + s(t-T)$ (D) Not related

23. $x(t) * u(t-t_0) =$

- (A) $\int_{-\infty}^{\infty} x(t) dt$ (B) $\int_{-\infty}^{t_0} x(t) dt$
 (C) $\int_{-\infty}^{t_0} x(\tau - t_0) d\tau$ (D) $\int_{-\infty}^{t-t_0} x(\tau) d\tau$

24. The range of value of a and b for stability of an LTI system $h[n] = (-3a)^n; n \geq 0$

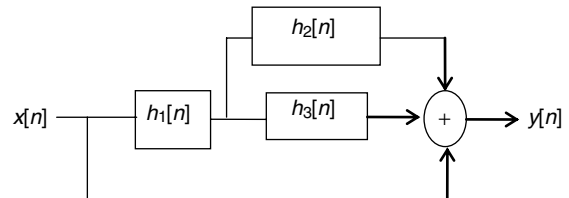
$$= b^{-n}, n < 0$$

- (A) $0 < |a| < \frac{1}{3}; 0 < |b| < 2$
 (B) $0 < |a| < \frac{1}{3}; 0 < |b| < 1$
 (C) $0 < |a| < \frac{1}{3}; 0 < |b| < 3$
 (D) $0 < |a| < \frac{1}{2}; 0 < |b| < 2$

25. The system $y(t) = 2\sin t + 3\cos t$ is

- (A) non-linear, time variant, with memory, causal
 (B) non-linear, time invariant, memoryless, causal
 (C) linear, time invariant, with memory, causal
 (D) linear, time variant, with memory, non-causal

26. The equivalent impulse response for this system



- (A) $(h_1[n] + h_3[n]) * (h_1[n] + h_2[n]) + 1$
 (B) $h_1[n] * (h_3[n] + h_2[n] + 1)$
 (C) $h_1[n] * (h_3[n] + h_2[n]) + 1$
 (D) $(h_3[n] + h_1[n]) * h_2[n] + 1$

27. The system $h[n] = \delta[n] + \delta[n+2] + \delta[n-2]$

- (A) Causal and stable
 (B) Causal but not stable
 (C) Not causal but stable
 (D) Neither causal nor stable

28. Given $x[n]$ defined in range $-6 \leq n \leq 3$, $y[n]$ in range $-2 \leq n \leq 7$, then the range of their convolution?

- (A) $-6 \leq n \leq 7$ (B) $-2 \leq n \leq 3$
 (C) $-8 \leq n \leq 7$ (D) $-8 \leq n \leq 10$

29. Response of a system to a complex input $x(t) = e^{j5t}$ is given as $y(t) = 5te^{j5t}$, then the system is

- (A) not LTI (B) LTI
 (C) may be LTI (D) not predictable

30. Impulse response of two LTI systems is given. $h_1(t) = e^{-(5-4j)t} u(t)$, $h_2(t) = e^{-3t} \cos 4tu(t)$ and the stable system is

- (A) Only $h_1(t)$ (B) Only $h_2(t)$
 (C) Both $h_1(t)$ and $h_2(t)$ (D) None of these

Practice Problems 2

Direction for questions 1 to 30: Select the correct alternative from the given choices.

1. Determine the fundamental period of these signals.

(1) $\cos \frac{\pi}{4} t + \sin \frac{\pi}{3} t$

(2) $\sin \sqrt{2} t + \cos 2t$

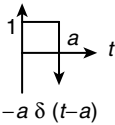
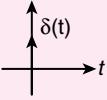
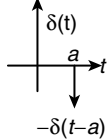
(A) 12, $\sqrt{2}$

(B) 12, $\frac{1}{\sqrt{2}}$

(C) 24, non-periodic

(D) 24π , non-periodic

2. Match the first order derivatives of group (1) to group (2)

Group (1)		Group (2)	
P	$u(t)$	1.	
Q	$u(t) - u(t-a)$ $a > 0$	2.	
R	$t u(t) - u(t-a)$ $a > 0$	3.	
S	$\text{Sgn}(t)$	4.	

(A) P - 1, Q - 4, R - 3, S - 2

(B) P - 3, Q - 4, R - 2, S - 1

(C) P - 3, Q - 2, R - 4, S - 1

(D) P - 3, Q - 1, R - 2, S - 4

3. Periodic signals are

(A) $f[n + mN] = f[n]$

(B) $f(t - T) = f(t)$

(C) $f(t + T) = f(t)$

(D) A, B, C

4. Which are correct regarding power and energy signals, respectively?

(A) $0 < P < \infty, E = \infty; 0 < E < \infty, P = 0$

(B) $0 < P < \infty, E = 0; 0 < E < \infty, P = \infty$

(C) $0 < P < \infty, E = 0; 0 < E < \infty, P = 0$

(D) $0 < P < \infty, E = \infty; 0 < E < \infty, P = \infty$

5. Find the wrong statement

(A) $\int_{-\infty}^{\infty} \delta(t) dt = 1$

(B) $\delta(-t) = \delta(t)$

(C) $\sin\left(t - \frac{\pi}{2}\right) \delta(t) = -1$

(D) $\cos t \cdot \delta(t - \pi) = -\delta(t - \pi)$

6. Discrete-time unit impulse can be obtained as

(A) $\delta[n] = \frac{du[n]}{dn}$

(B) $\delta[n] = u[n] - u[n-1]$

(C) A or B

(D) $\delta[n] = \int_{-\infty}^{\infty} u[n] dn$

7. For unit impulse signal of zero duration, the magnitude is

(A) 1

(B) ∞

(C) 0

(D) t

8. Match the following

Group (1)	Group (2)
P. $\int_{-\infty}^{\infty} \delta(t) dt$	1. 0
Q. $\int_{-\infty}^{\infty} x(\lambda) \cdot \delta(t - \lambda) d\lambda$	2. 1
R. $\int_{-\infty}^{\infty} x(t) \delta(t - \lambda) dt$	3. $\frac{1}{ a } \delta(t)$
S. $\delta(at)$	4. $ a \delta(t)$
	5. $x(t)$
	6. $x(\lambda)$

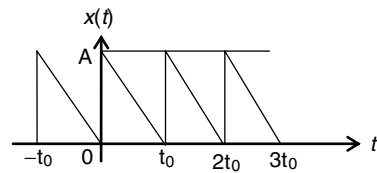
(A) P - 2, Q - 6, R - 5, S - 4

(B) P - 2, Q - 5, R - 6, S - 3

(C) P - 1, Q - 5, R - 6, S - 4

(D) P - 1, Q - 6, R - 5, S - 3

9. Find the value of $x(0)$ for the following sawtooth wave shown in the following figure.



(A) A

(B) $A/2$

(C) Undefined

(D) 0

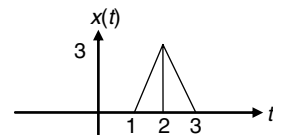
10. The energy of signal is

(A) 3 joules

(B) 9 joules

(C) 6 joules

(D) 12 joules



11. Find the mutually orthogonal functions below.

(1) $\sin m\omega_0 t, \cos m\omega_0 t$

(2) $\sin m\omega_0 t, \sin n\omega_0 t$

(3) $\sin m\omega_0 t, \tan n\omega_0 t$

(4) $\cos n\omega_0 t, \cos m\omega_0 t$

(5) $\tan n\omega_0 t, \sec m\omega_0 t$

(A) 1, 2, 4

(B) 1, 2, 4, 5

(C) 2, 4

(D) 1, 3, 5

12. Two complex functions $f_1(t)$ and $f_2(t)$ are orthogonal over the interval (t_1, t_2) if

(A) $\int_{t_1}^{t_2} f_1(t) f_2^*(t) dt = 0$ (B) $\int_{t_1}^{t_2} f_1^*(t) f_2(t) dt = 0$

(C) Both A and B (D) $\int_{t_1}^{t_2} f_1(t) * f_2(t) dt = 0$

13. Which of the following is not correct about the signals $e^{j\omega t}$, $e^{-j\omega t}$?

- (1) Both oscillate at same frequency ω
 (2) One oscillate at ' ω ' frequency, other at ' $-\omega$ '.
 (3) Two phasors rotating in opposite directions.
 (4) When added, they yield a real function.

- (A) 2, 4 (B) 1, 3, 4
 (C) 4 (D) 2

14. If $y(t)$ is the output of a CT – LTI system with input $x(t)$,

the output of the system if the input is $\frac{dx(t)}{dt}$

- (A) $-ty(t)$ (B) $ty(t)$
 (C) $\frac{d}{dt} y(t)$ (D) $\int y(t) dt$

15. If $h(t)$ is the impulse response of causal, linear, time-invariant continuous system, then the output of the system for input $x(t)$ is

(A) $\int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$ (B) $\int_0^t x(\tau) h(t - \tau) d\tau$
 (C) $\int_{-\infty}^0 x(\tau) h(t - \tau) d\tau$ (D) $\int_0^{\infty} x(\tau) h(t - \tau) d\tau$

16. Which of the following system is time invariant?

- (A) $y[n] = x[n] - x[n - 1]$
 (B) $y[n] = n x[n]$
 (C) $y[n] = x[-n]$
 (D) $y[n] = n x^2[n]$

17. Which of the following system is time variant?

- (A) $y[n] = x[n] - x[n - 1]$
 (B) $y[n] = x[n] - 6x[n - 1]$
 (C) $y[n] = a^{x[n]}$
 (D) $y[n] = nx^2[n]$

18. Which of the following system is linear?

- (A) $y[n] = x^2[n]$ (B) $y[n] = Bx[n] + C$
 (C) $y[n] = nx^2[n]$ (D) $y[n] = nx[n]$

19. Which of the following system is causal?

- (A) $y[n] = x[n] + 3x[n + 4]$
 (B) $y[n] = x[n] - x[n - 1]$
 (C) $y[n] = x[n^2]$
 (D) $y[n] = x[2n]$

20. Pick up the non-causal system?

(A) $y[n] = \sum_{m=-\infty}^n x[n]$ (B) $y[n] = ax[n]$

(C) $y[n] = x[-n]$ (D) $y[n] = nx[n]$

21. The response of the LTI system whose input $x[n] = \{-1, 2, 3, 1\}$ impulse response

$$h[n] = \{1, 2, 1, -1\}$$

(A) $\{-1, 0, 6, 10, 3, -2, -1\}$

(B) $\{1, 4, 8, 8, 3, -2, -1\}$

(C) $\{-1, 0, 6, 10, 3, -2, -1\}$

(D) $\{1, 4, 8, 8, 3, -2, -1\}$

22. The discrete-time system

$$y[n] = x[n - 3] - 4x[n - 7]$$
 is a

- (A) dynamic system (B) time-varying system
 (C) memoryless system (D) non-linear system

23. Which of the following system is causal?

(A) $h[n] = n \left(\frac{1}{2}\right)^{n+1} u[n + 1]$

(B) $y[n] = x^2[n] - x^2(n + 1)$

(C) $y[n] = x[-n] + x[2n + 1]$

(D) $h[n] = n \left(\frac{1}{2}\right)^{n+1} u[n]$

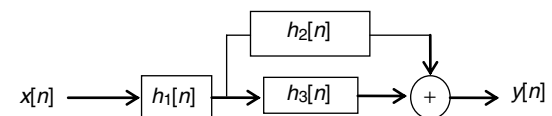
24. The system $y[n] = \sqrt{x[n]} + 3\sqrt{x[n + 3]}$

- (A) non-linear, time variant, memory, causal, stable
 (B) linear, time invariant, memoryless, causal unstable
 (C) non-linear, time invariant, memoryless, non-causal, unstable
 (D) non-linear, time invariant, memory, non-causal, stable

25. $x(t) = u(t)$ is

- (A) Power signal with power = -1
 (B) Energy signal with energy = 1
 (C) Power signal with power = 1
 (D) Energy signal with energy = -1

26. The equivalent impulse response of the system



(A) $(h_1[n] + h_3[n]) * (h_1[n] + h_2[n])$

(B) $(h_1[n] * h_3[n]) + h_1[n] * h_2[n]$

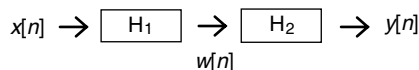
(C) $(h_1[n] + h_3[n]) * h_2[n]$

(D) $(h_1[n] + h_2[n]) * h_3[n]$

27. The period of the $x[n] = \cos \frac{\pi n}{4} \cos \frac{\pi n}{3}$ is

- (A) 12 (B) 24
 (C) 6 (D) not periodic

28. Consider the cascade of following two systems H_1 and H_2



$$H_1 \text{ is given by } w(n) = \frac{1}{4}w(n-1) + x[n]$$

$$H_2 \text{ is given by } y[n] = ay[n-1] + bw[n]$$

The difference equation for cascaded system is $y[n] +$

$$\frac{1}{4}y[n-2] - \frac{5}{4}y[n-1] = x[n]$$

Then, the value of 'a' is

- (A) 1 (B) 1/3 (C) 1/4 (D) 1/2

29. The value of b is
(A) 1 (B) 1/2 (C) 1/3 (D) 1/4

30. Impulse response of
 $y[n] = x[n] + 2x[n-1] - 4 \times [n-2]$

- (A) $\delta[n] - 2\delta[n-1]$
(B) $\delta[n] + 2\delta[n-1]$
(C) $\delta[n] + 2\delta[n-1] + 4\delta[n-2]$
(D) $\delta[n] + 2\delta[n-1] - 4\delta[n-2]$

PREVIOUS YEARS' QUESTIONS

1. The impulse response $h[n]$ of a linear time-invariant system is given by $h[n] = u[n+3] + u[n-2] - 2u[n-7]$, where $u[n]$ is the unit-step sequence. The abovementioned system is [2004]

- (A) stable but not causal
(B) stable and causal
(C) causal but unstable
(D) unstable and not causal

2. The impulse response $h[n]$ of linear time-invariant system is given as

$$h[n] = \begin{cases} -2\sqrt{2} & n = 1, -1 \\ 4\sqrt{2} & n = 2, -2 \\ 0 & \text{otherwise} \end{cases}$$

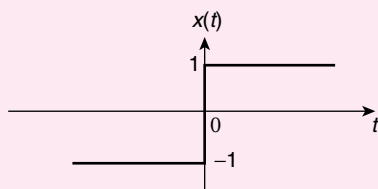
If the input to the abovementioned system is the

sequence $e^{\frac{j\pi n}{4}}$, then the output is [2004]

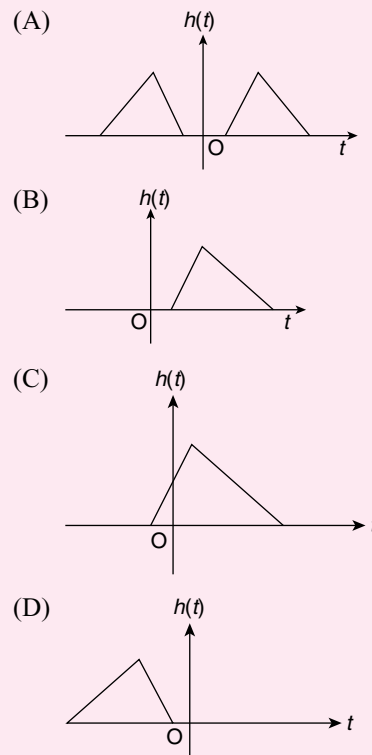
- (A) $4\sqrt{2}e^{\frac{j\pi n}{4}}$ (B) $4\sqrt{2}e^{-\frac{j\pi n}{4}}$
(C) $4e^{\frac{j\pi n}{4}}$ (D) $-4e^{\frac{j\pi n}{4}}$

3. The function $x(t)$ is shown in the figure. Even and odd parts of a unit-step function $u(t)$ are, respectively. [2005]

- (A) $\frac{1}{2}, \frac{1}{2}x(t)$ (B) $-\frac{1}{2}, \frac{1}{2}x(t)$
(C) $\frac{1}{2}, -\frac{1}{2}x(t)$ (D) $-\frac{1}{2}, -\frac{1}{2}x(t)$



4. Which of the following can be impulse response of a causal system? [2005]



5. The power in the signal
 $s(t) = 8\cos\left(20\pi t - \frac{\pi}{2}\right) + 4\sin(15\pi t)$ is: [2005]

- (A) 40 (B) 41 (C) 42 (D) 82

6. The Dirac delta function $\delta(t)$ is defined as [2006]

- (A) $\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & \text{otherwise} \end{cases}$
(B) $\delta(t) = \begin{cases} \infty & t = 0 \\ 0 & \text{otherwise} \end{cases}$

$$(C) \quad \delta(t) = \begin{cases} 1 & t = 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$(D) \quad \delta(t) = \begin{cases} \infty & t = 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t) dt = 1$$

7. A system with input $x[n]$ and output $Y[n]$ is given as

$$y[n] = \left(\sin \frac{5}{6} \pi n \right) x(n). \quad \text{The system is} \quad [2006]$$

- (A) linear, stable, and invertible
 (B) non-linear, stable, and non-invertible
 (C) linear, stable, and non-invertible
 (D) linear, unstable, and invertible

8. A Hilbert transformer is a [2007]

- (A) non-linear system (B) non-causal system
 (C) time-varying system (D) low-pass system

9. The input and output of a continuous time system are, respectively, denoted by $x(t)$ and $y(t)$. Which of the following descriptions correspond to a causal system? [2008]

- (A) $y(t) = x(t-2) + x(t+4)$
 (B) $y(t) = (t-4)x(t+1)$
 (C) $y(t) = (t+4)x(t-1)$
 (D) $y(t) = (t+5)x(t+5)$

10. The impulse response $h(t)$ of a linear time-invariant continuous time system is described by $h(t) = \exp(\alpha t)u(t) + \exp(\beta t)u(-t)$, where $u(t)$ denotes the unit-step function, and α and β are real constants. This system is stable if [2008]

- (A) α is positive and β is positive
 (B) α is negative and β is negative
 (C) α is positive and β is negative
 (D) α is negative and β is positive

11. A discrete time-linear shift-invariant system has an impulse response $h[n]$ with $h[0] = 1$, $h[1] = -1$, $h[2] = 2$, and 0, otherwise. The system is given an input sequence $x[n]$ with $x[0] = x[2] = 1$, and zero, otherwise. The number of non-zero samples in the output sequence $y[n]$, and the value of $y[2]$ are, respectively, [2008]
 (A) 5, 2 (B) 6, 2 (C) 6, 1 (D) 5, 3

12. Let $x(t)$ be the input and $y(t)$ be the output of a continuous time system. Match the system properties P_1 , P_2 , and P_3 with system relations R_1 , R_2 , R_3 , and R_4 .

Properties	Relations
P_1 : Linear but NOT time-invariant	$R_1: y(t) = t^2 x(t)$
P_2 : Time-invariant but NOT linear	$R_2: y(t) = t x(t) $
P_3 : Linear and time-invariant	$R_3: y(t) = x(t) $
	$R_4: y(t) = x(t-5)$

[2008]

- (A) $(P_1, R_1), (P_2, R_3), (P_3, R_4)$
 (B) $(P_1, R_2), (P_2, R_3), (P_3, R_4)$
 (C) $(P_1, R_3), (P_2, R_1), (P_3, R_2)$
 (D) $(P_1, R_1), (P_2, R_2), (P_3, R_3)$

13. A function is given $f(t) = \sin^2 t + \cos 2t$. Which of the following is true? [2009]

- (A) f has frequency components at 0 and $\frac{1}{2}\pi$ Hz
 (B) f has frequency components at 0 and $1/\pi$ Hz
 (C) f has frequency components at $\frac{1}{2}\pi$ and $1/\pi$ Hz
 (D) f has frequency components at 0, $\frac{1}{2}\pi$ and $1/\pi$ Hz

14. Consider a system whose input x and output y are related by the equation

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(2\tau)d\tau,$$

where $h(t)$ is shown in the graph.

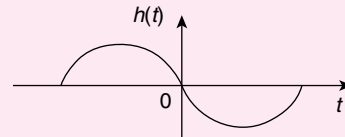
Which of the following four properties are possessed by the system?

BIBO: Bounded input gives a bounded output

Causal: The system is causal

LP: The system is low pass

LTI: The system is linear and time invariant [2009]



- (A) Causal, LP (B) BIBO, LTI
 (C) BIBO, Causal, LTI (D) LP, LTI

15. Two discrete-time systems with impulse responses $h_1[n] = \delta[n-1]$ and $h_2[n] = \delta[n-2]$ are connected in cascade. The overall impulse response of the cascaded system is [2010]

- (A) $\delta[n-1] + \delta[n-2]$ (B) $\delta[n-4]$
 (C) $\delta[n-3]$ (D) $\delta[n-1]\delta[n-2]$

16. A system is defined by its impulse response $h(n) = 2^n u(n-2)$. The system is [2011]

- (A) stable and causal
 (B) causal but not stable
 (C) stable but not causal
 (D) unstable and non-causal

17. If the unit step response of a network is $(1 - e^{-\alpha t})$, then its unit-impulse response is [2011]

- (A) $\alpha e^{-\alpha t}$ (B) $\alpha^{-1} e^{-\alpha t}$
 (C) $(1 - \alpha^{-1})e^{-\alpha t}$ (D) $(1 - \alpha)e^{-\alpha t}$

18. The input $x(t)$ and output $y(t)$ of a system are related as $y(t) = \int_{-\infty}^1 x(\tau) \cos(3\tau) d\tau$. The system is [2012]

- (A) time invariant and stable
 (B) stable and not time invariant
 (C) time invariant and not stable
 (D) not time invariant and not stable

19. Two systems with impulse responses $h_1(t)$ and $h_2(t)$ are connected in cascade. Then, the overall impulse response of the cascaded system is given by [2013]

(A) product of $h_1(t)$ and $h_2(t)$
 (B) sum of $h_1(t)$ and $h_2(t)$
 (C) convolution of $h_1(t)$ and $h_2(t)$
 (D) subtraction of $h_2(t)$ from $h_1(t)$

20. For a periodic signal $V(t) = 30 \sin 100t + 10 \cos 300t + 6 \sin(500t + \pi/4)$, the fundamental frequency in rad/s is [2013]

(A) 100 (B) 300
 (C) 500 (D) 1,500

21. The impulse response of a continuous time system is given by $h(t) = \delta(t - 1) + \delta(t - 3)$. The value of the step response at $t = 2$ is [2013]

(A) 0 (B) 1
 (C) 2 (D) 3

22. A discrete-time signal $x[n] = \sin(\pi^2 n)$, n being an integer, is [2014]

(A) periodic with period π
 (B) periodic with period π^2
 (C) periodic with period $\pi/2$
 (D) not periodic

23. Consider a discrete-time signal

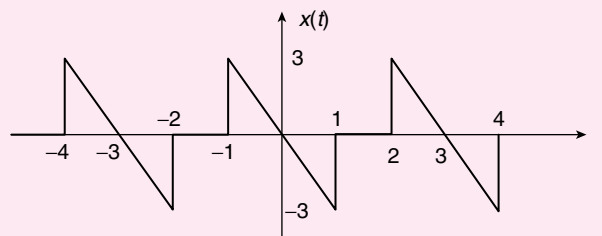
$$x[n] = \begin{cases} n & \text{for } 0 \leq n \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

If $y[n]$ is the convolution of $x[n]$ with itself, the value of $y[4]$ is [2014]

24. The result of the convolution $x(-t) \times \delta(-t - t_0)$ is [2015]

(A) $x(t + t_0)$ (B) $x(t - t_0)$
 (C) $x(-t + t_0)$ (D) $x(-t - t_0)$

25. The waveform of a periodic signal $x(t)$ is shown in the figure.



A signal $g(t)$ is defined by $g(t) = x\left(\frac{t-1}{2}\right)$. The average power of $g(t)$ is [2015]

26. The impulse response of an LTI system can be obtained by [2015]

(A) differentiating the unit ramp response
 (B) differentiating the unit-step response
 (C) integrating the unit ramp response
 (D) integrating the unit-step response

27. A continuous time function $x(t)$ is periodic with period T . The function is sampled uniformly with a sampling period T_s . In which one of the following cases is the sampled signal periodic? [2016]

(A) $T = \sqrt{2} T_s$ (B) $T = 1.2 T_s$
 (C) Always (D) Never

28. The energy of the signal $x(t) = \frac{\sin(4\pi t)}{4\pi t}$ is [2016]

ANSWER KEYS

EXERCISES

Practice Problems 1

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. B | 3. A | 4. A | 5. D | 6. A | 7. A | 8. B | 9. A | 10. C |
| 11. A | 12. B | 13. B | 14. B | 15. B | 16. D | 17. A | 18. A | 19. C | 20. B |
| 21. A | 22. B | 23. D | 24. B | 25. B | 26. C | 27. C | 28. D | 29. D | 30. C |

Practice Problems 2

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. D | 4. A | 5. C | 6. B | 7. B | 8. B | 9. B | 10. C |
| 11. A | 12. C | 13. D | 14. C | 15. B | 16. A | 17. D | 18. D | 19. B | 20. C |
| 21. A | 22. A | 23. D | 24. D | 25. C | 26. B | 27. B | 28. A | 29. A | 30. D |

Previous Years' Questions

- | | | | | | | | | | |
|-------|-------|-----------------|-------|-------|-------|-------|----------|-------|-------|
| 1. A | 2. D | 3. A | 4. B | 5. A | 6. D | 7. C | 8. B | 9. C | 10. D |
| 11. D | 12. A | 13. B | 14. B | 15. C | 16. B | 17. A | 18. D | 19. C | 20. A |
| 21. B | 22. D | 23. 9.9 to 10.1 | 24. D | 25. 2 | 26. B | 27. B | 28. 0.25 | | |